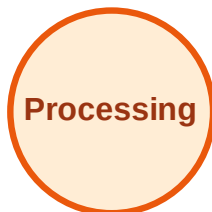


BFS Traversal

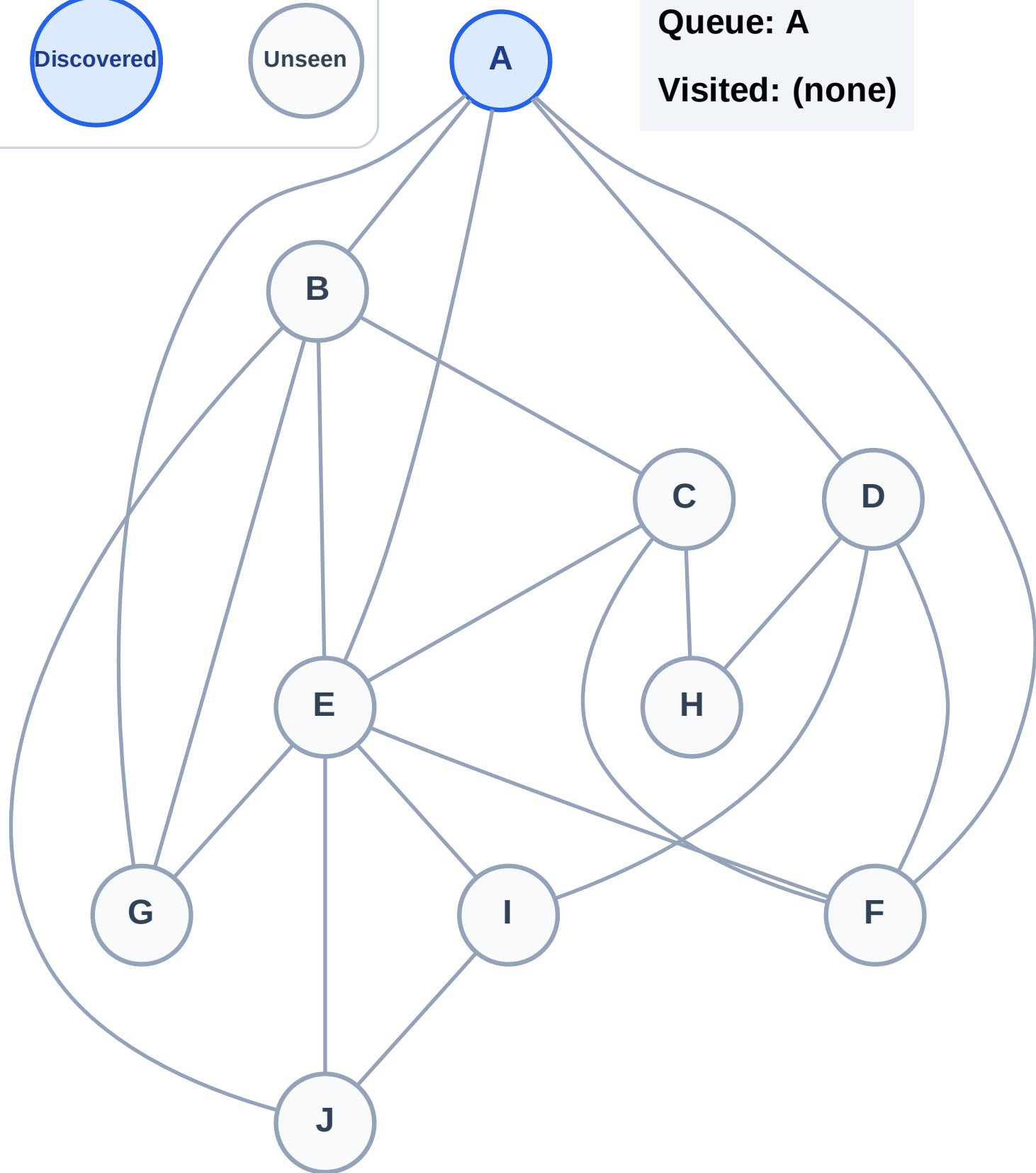
Step 1: Start at A

Legend



Queue: A

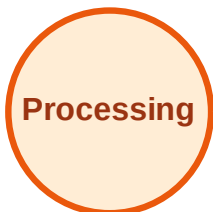
Visited: (none)



BFS Traversal

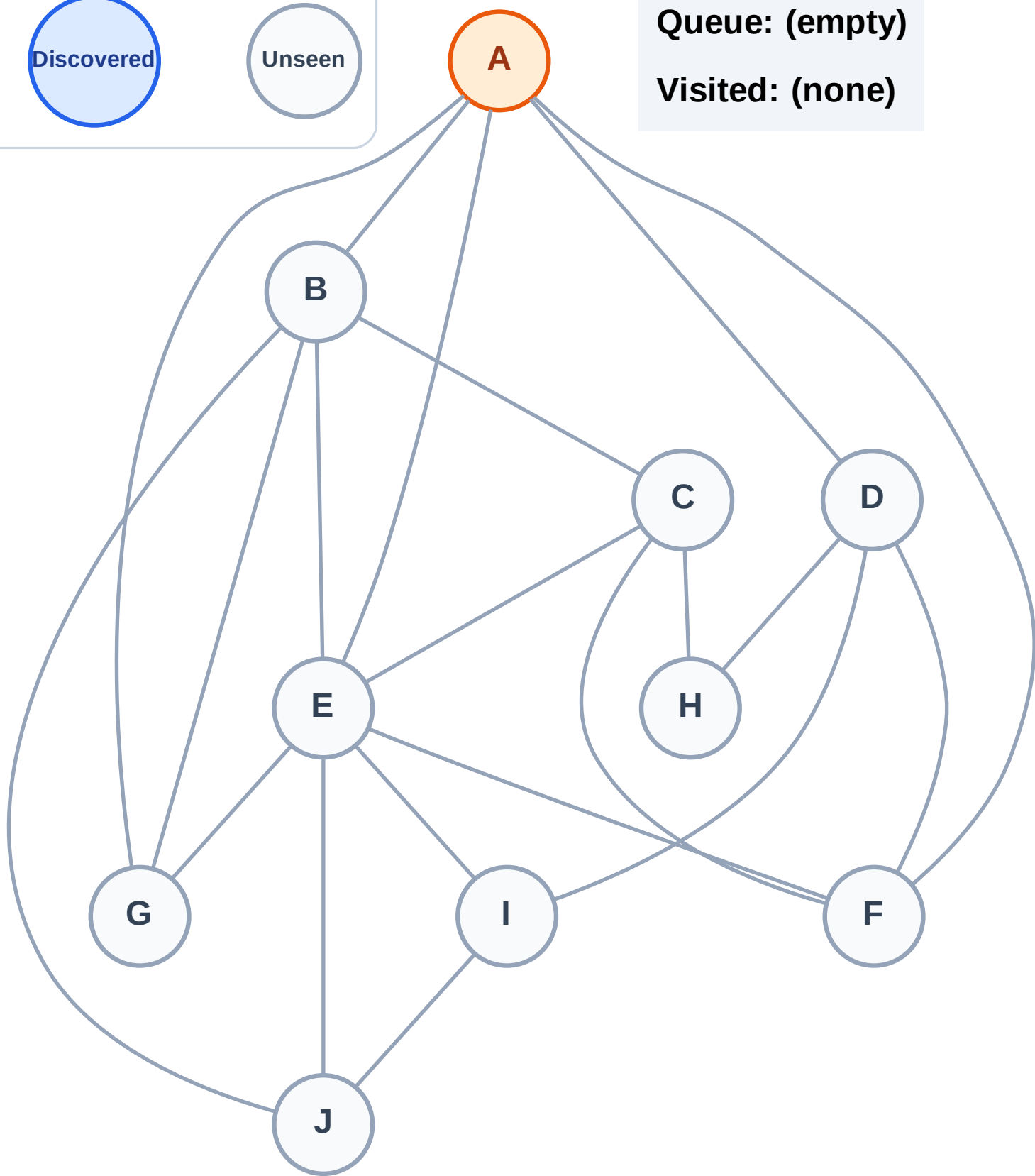
Step 2: Process node A

Legend



Queue: (empty)

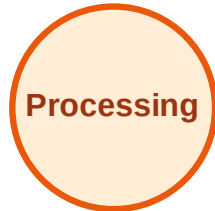
Visited: (none)



BFS Traversal

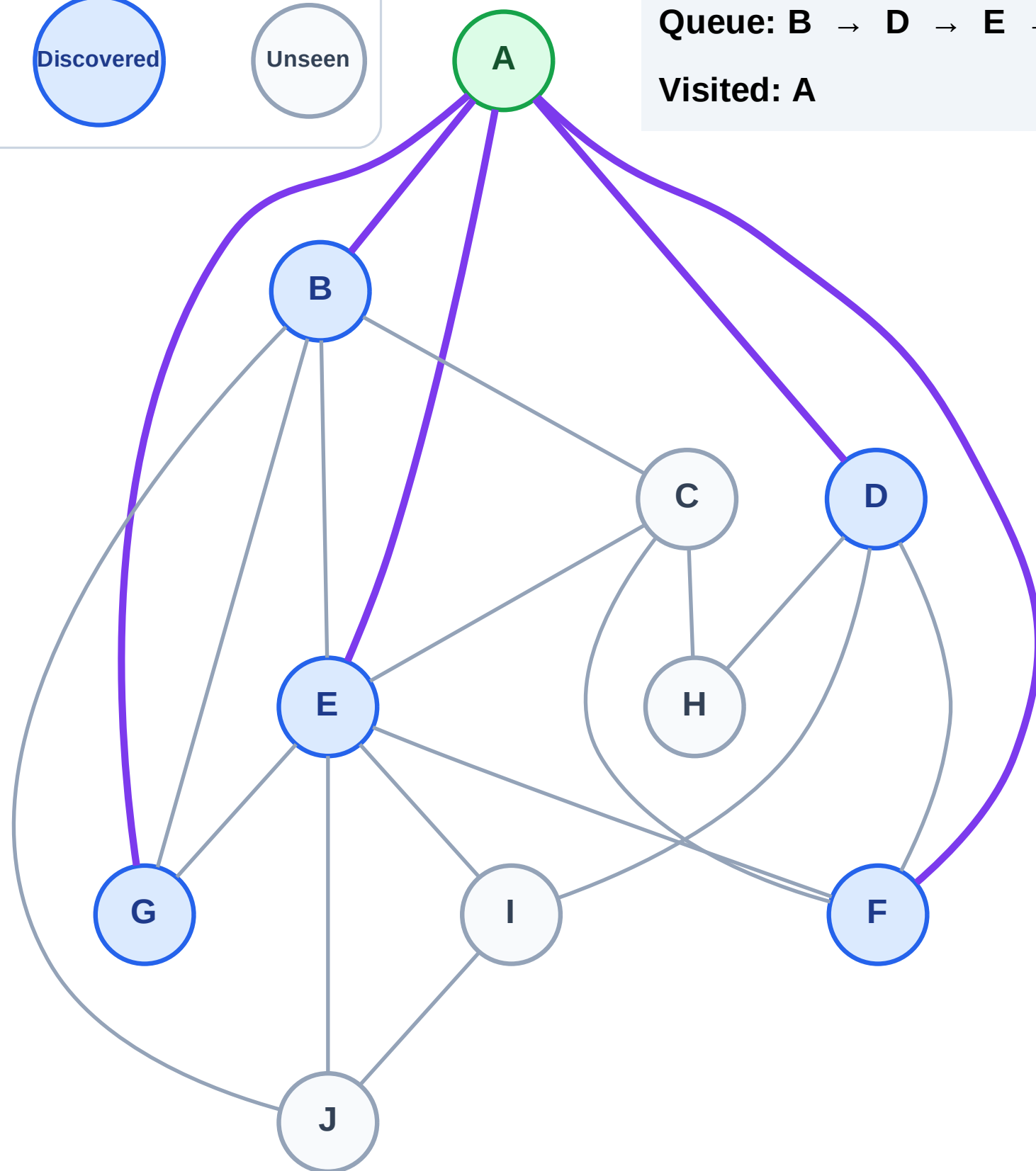
Step 3: Discover B, D, E, F, G from A

Legend



Queue: B → D → E → F → G

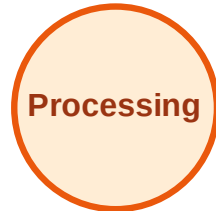
Visited: A



BFS Traversal

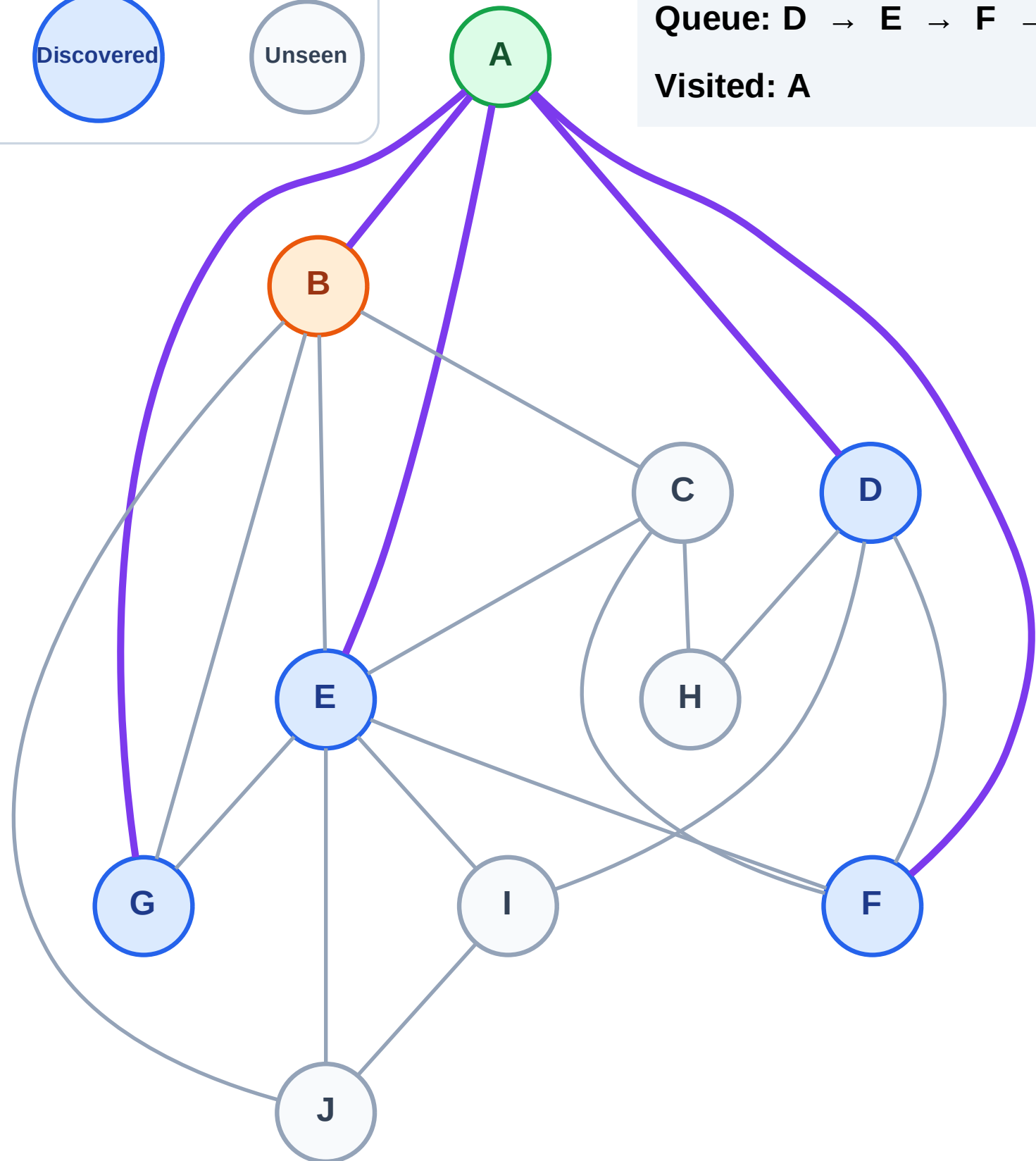
Step 4: Process node B

Legend



Queue: D → E → F → G

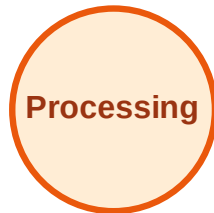
Visited: A



BFS Traversal

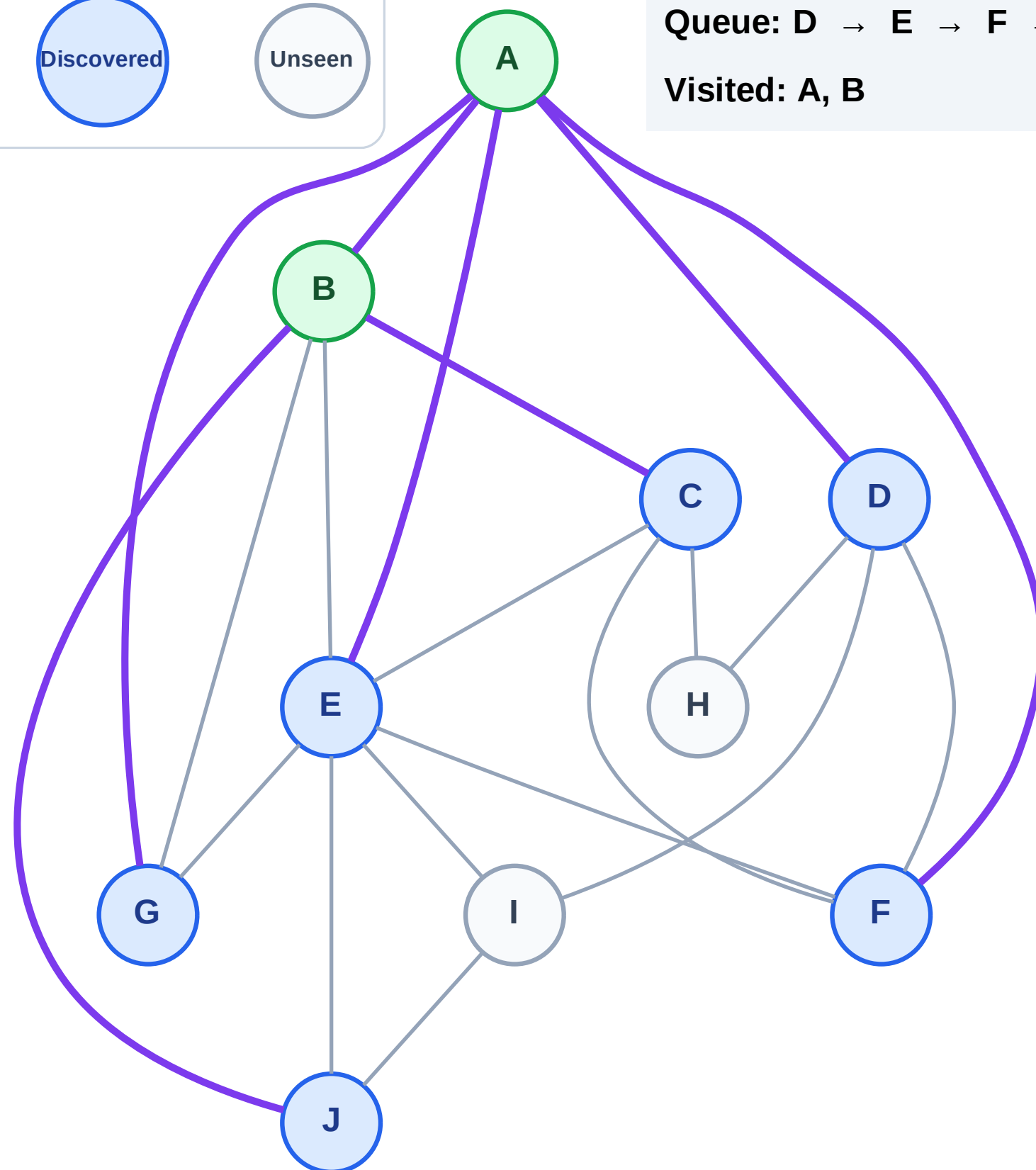
Step 5: Discover C, J from B

Legend



Queue: D → E → F → G → C → J

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

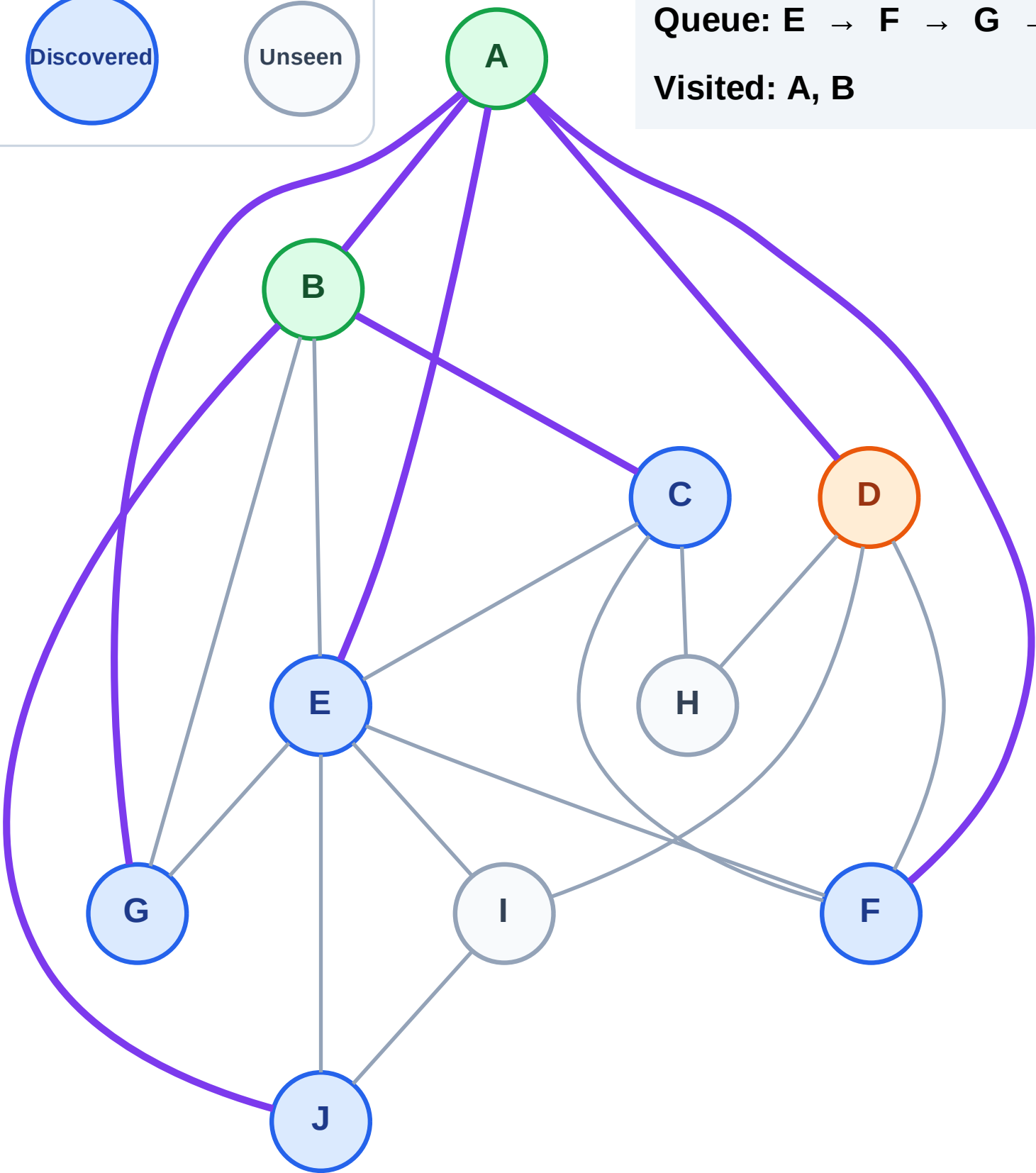
Processing

Discovered

Unseen

Queue: E → F → G → C → J

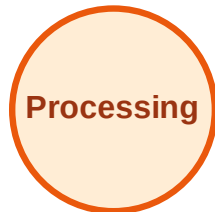
Visited: A, B



BFS Traversal

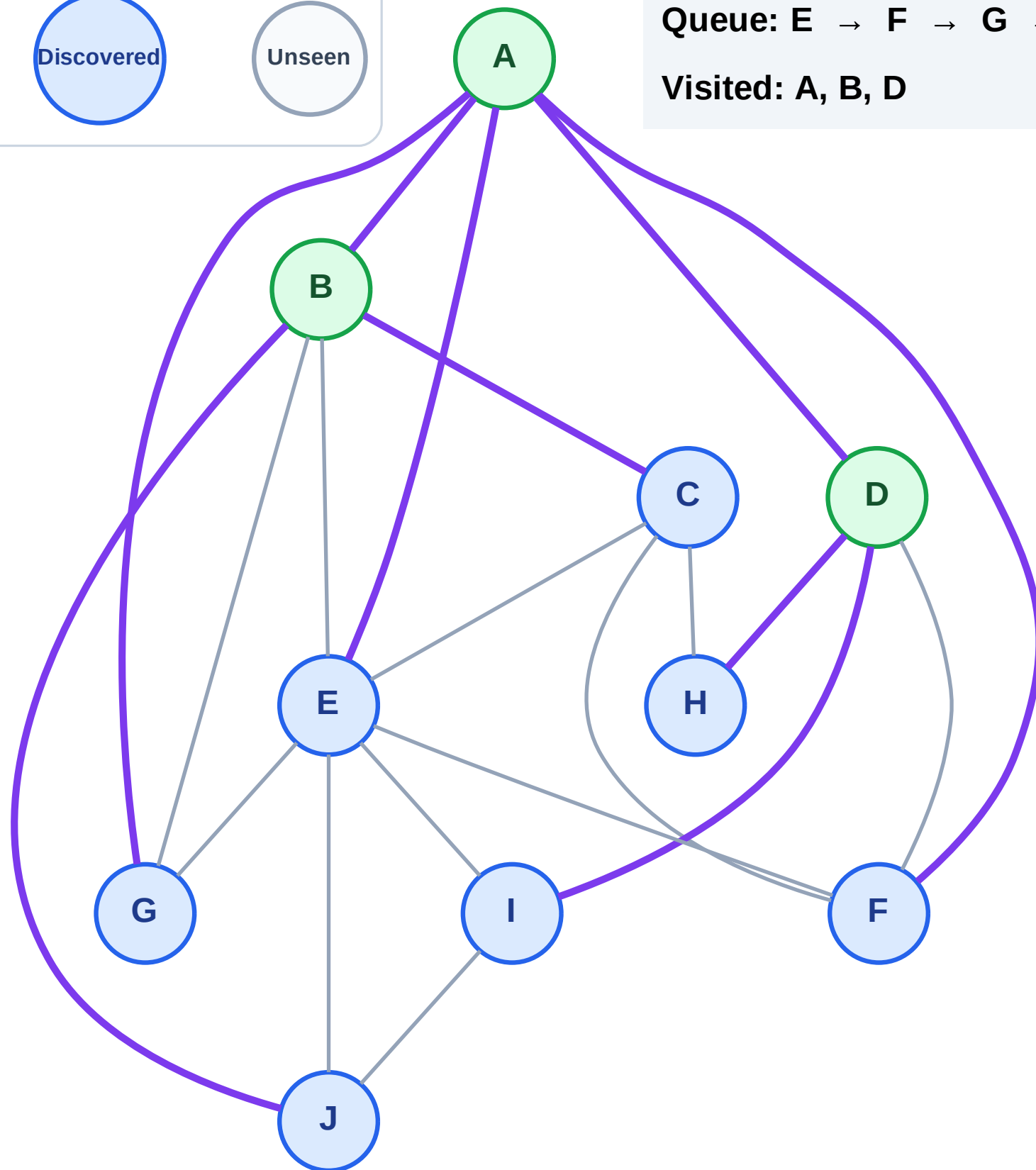
Step 7: Discover H, I from D

Legend



Queue: E → F → G → C → J → H → I

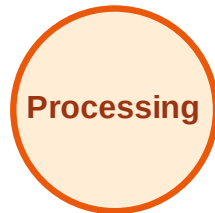
Visited: A, B, D



BFS Traversal

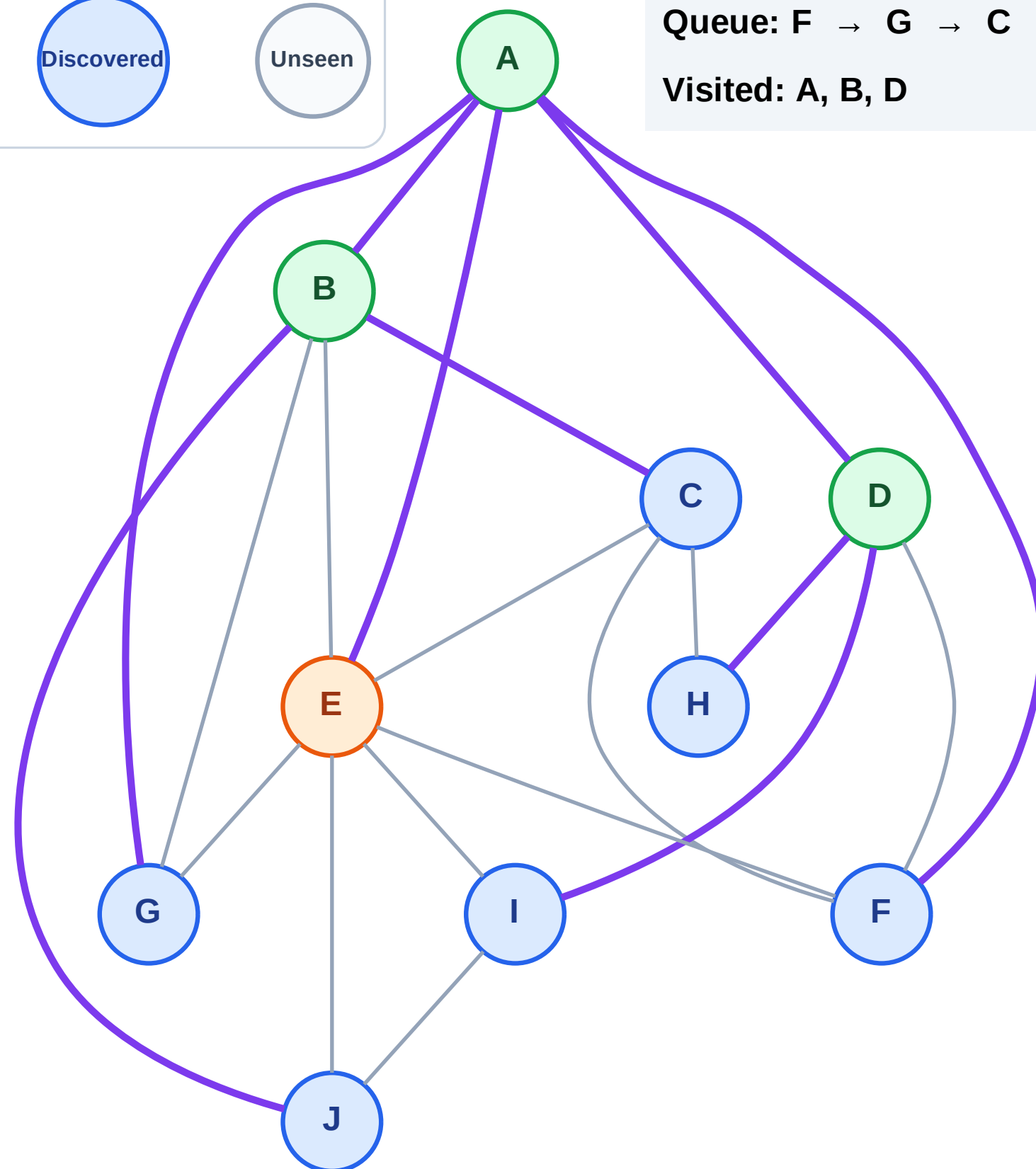
Step 8: Process node E

Legend



Queue: F → G → C → J → H → I

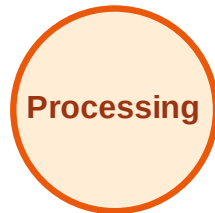
Visited: A, B, D



BFS Traversal

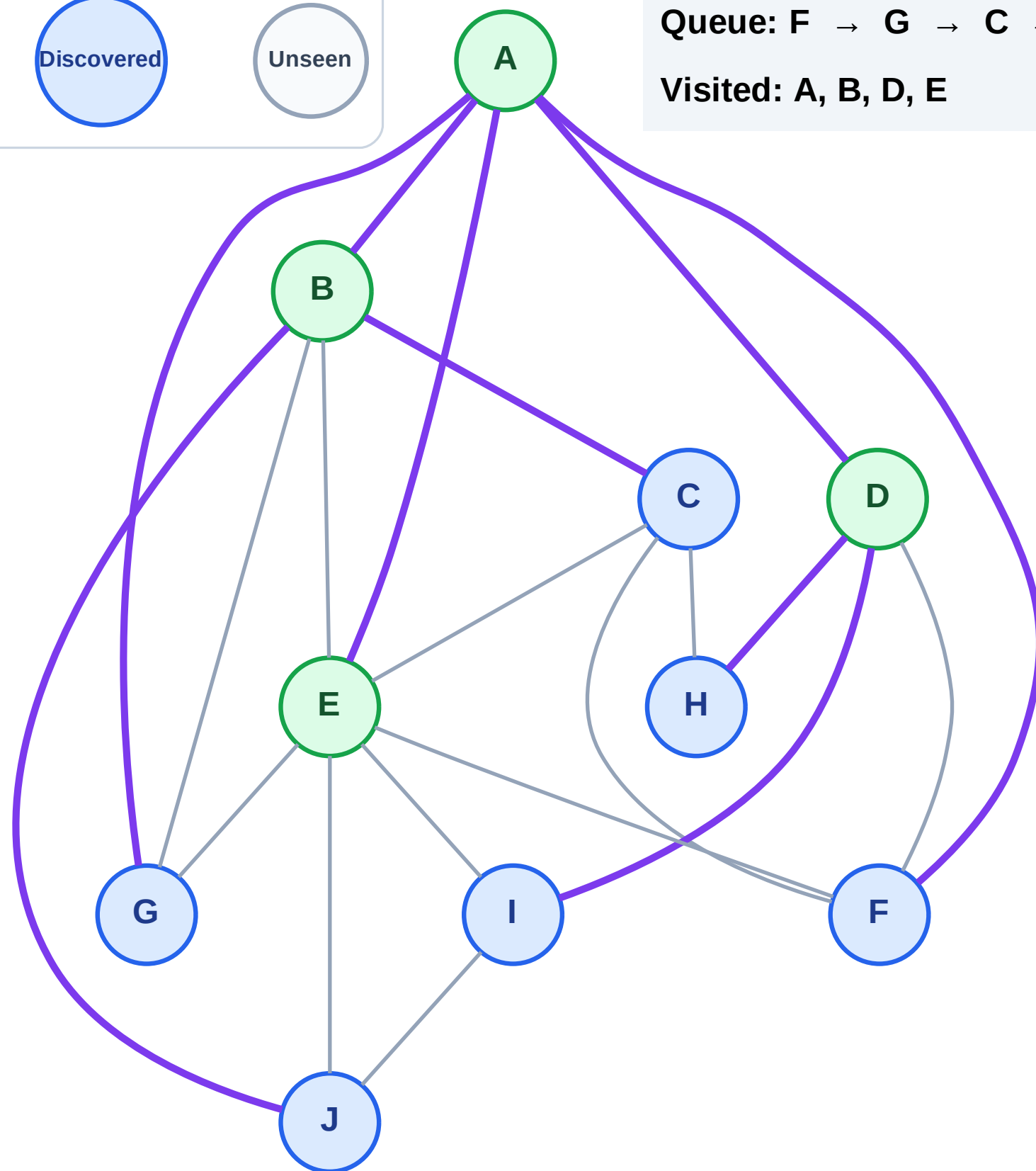
Step 9: No new neighbors from E

Legend



Queue: F → G → C → J → H → I

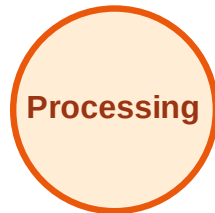
Visited: A, B, D, E



BFS Traversal

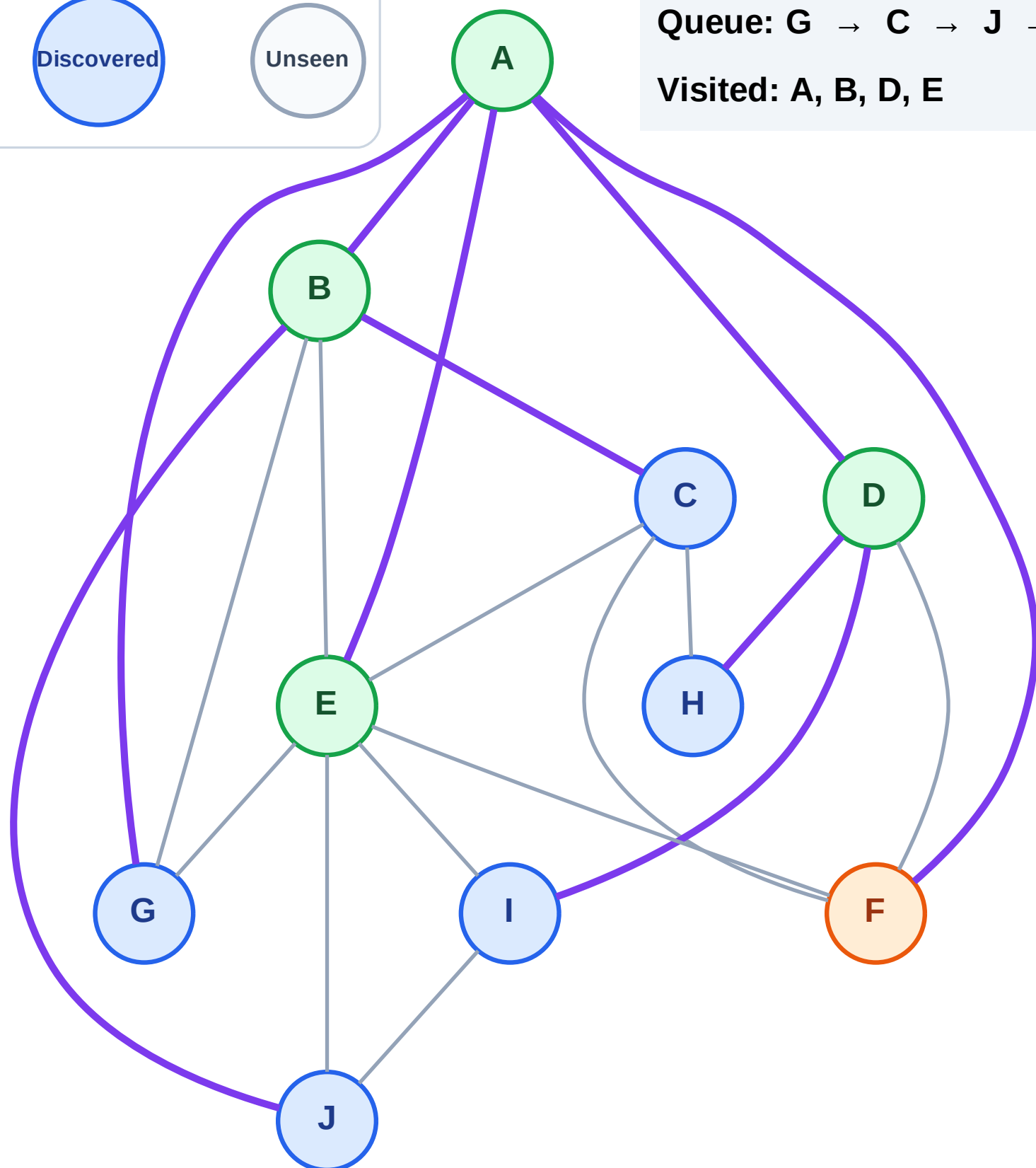
Step 10: Process node F

Legend



Queue: G → C → J → H → I

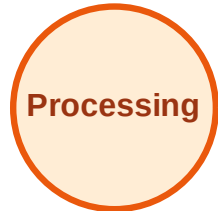
Visited: A, B, D, E



BFS Traversal

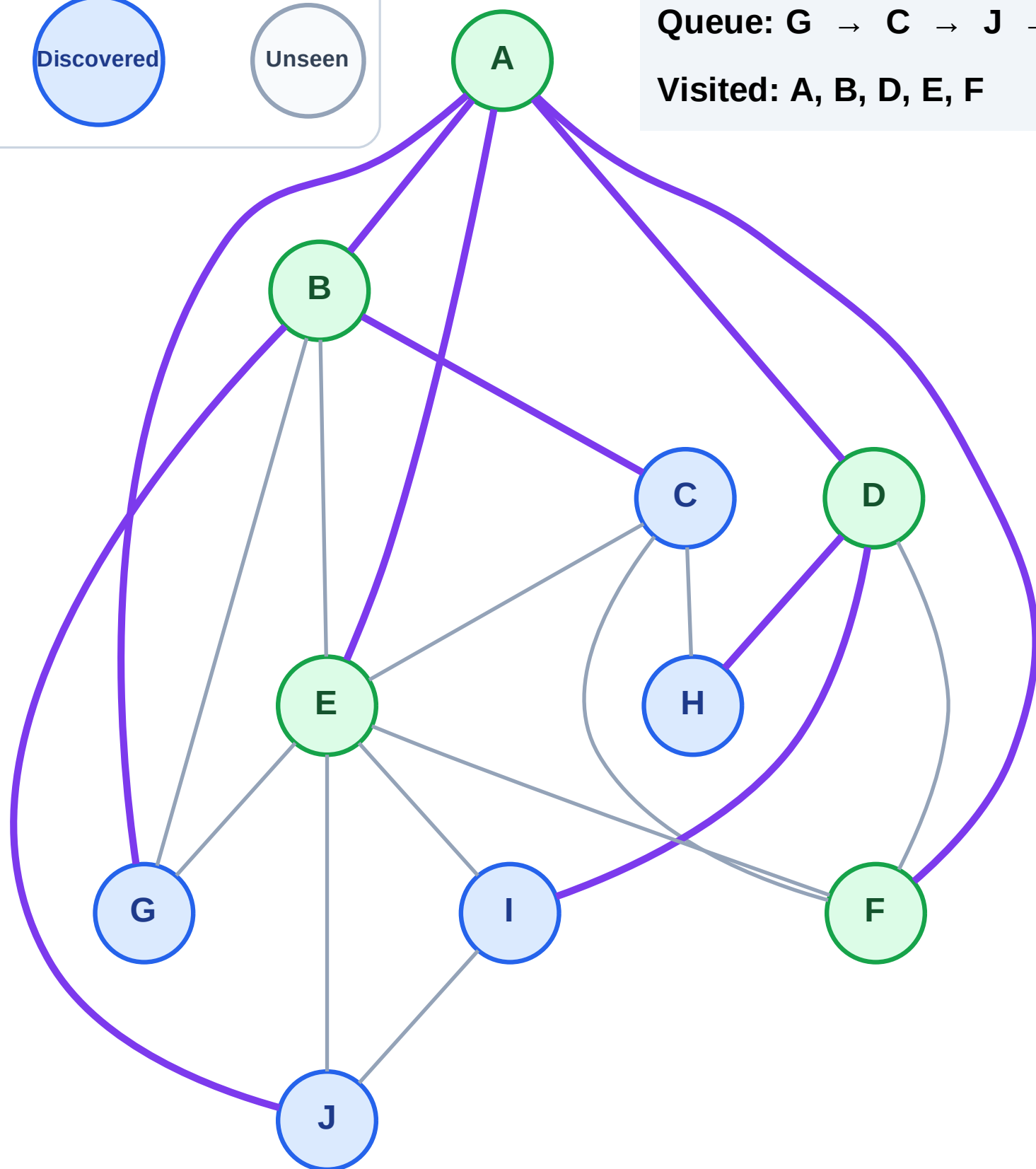
Step 11: No new neighbors from F

Legend



Queue: G → C → J → H → I

Visited: A, B, D, E, F



BFS Traversal

Step 12: Process node G

Legend

Complete

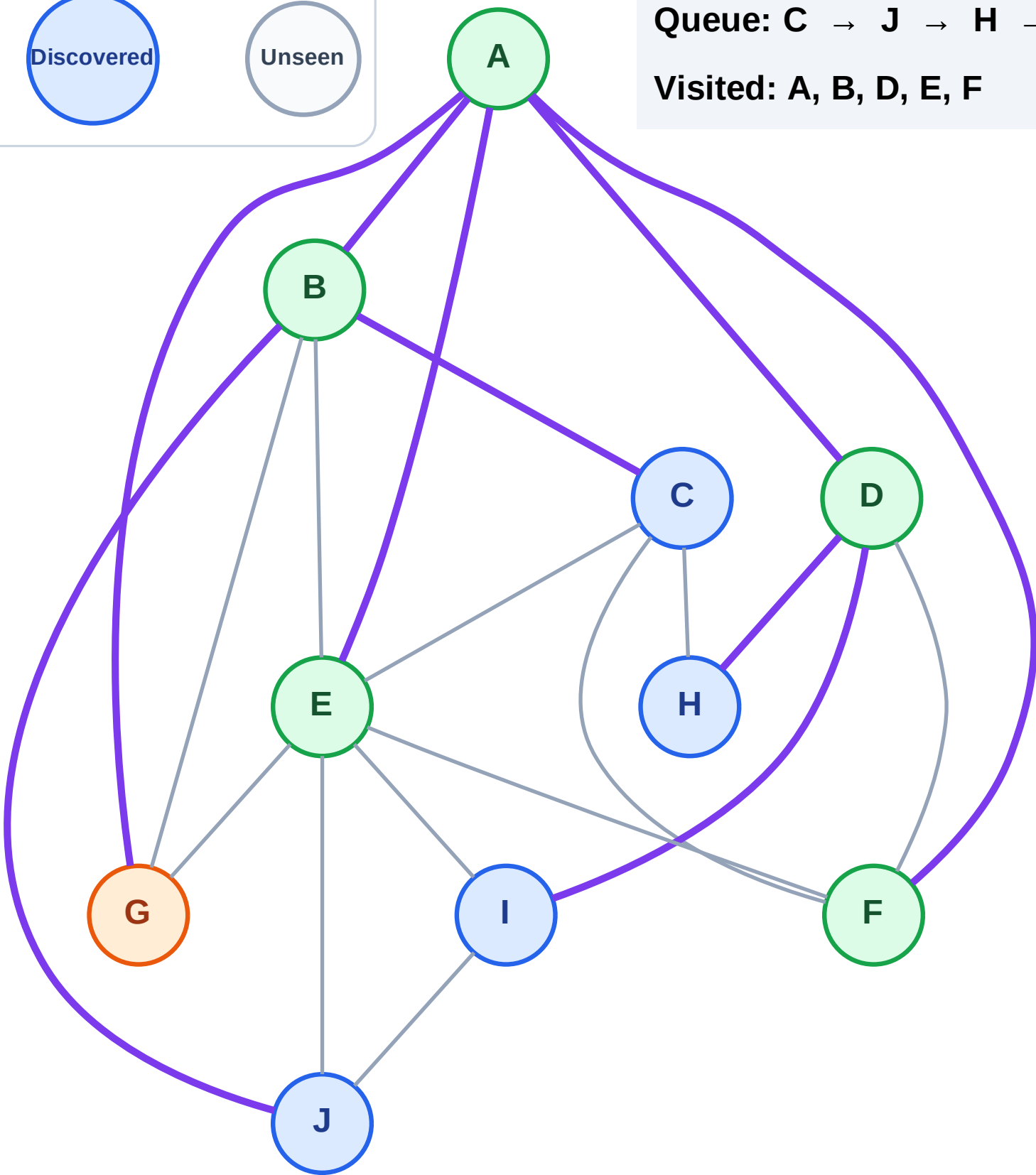
Processing

Discovered

Unseen

Queue: C → J → H → I

Visited: A, B, D, E, F



BFS Traversal

Step 13: No new neighbors from G

Legend

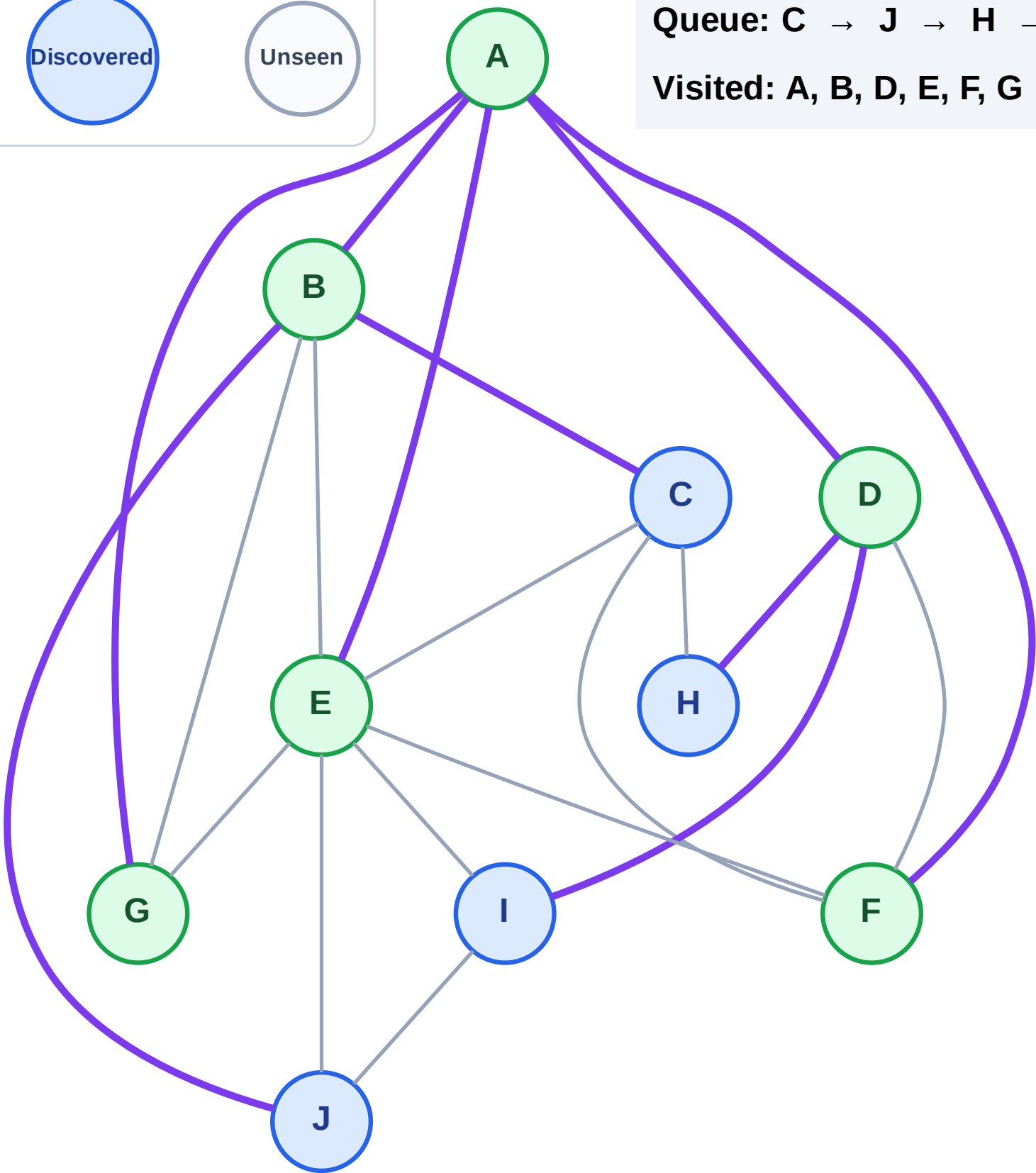
Complete

Processing

Discovered

Unseen

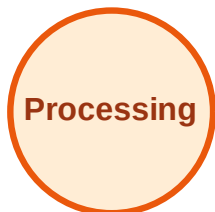
Queue: C → J → H → I
Visited: A, B, D, E, F, G



BFS Traversal

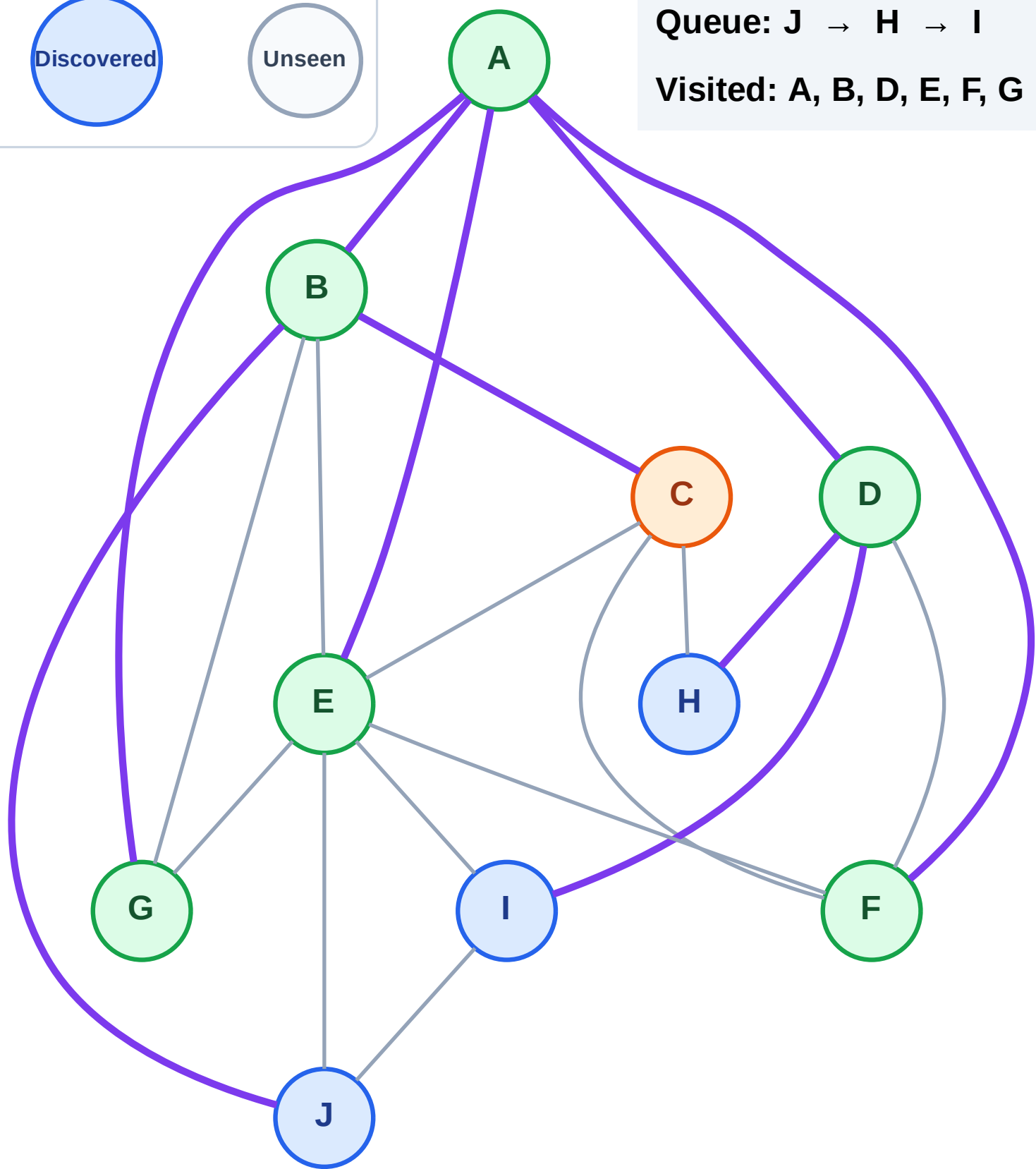
Step 14: Process node C

Legend



Queue: J → H → I

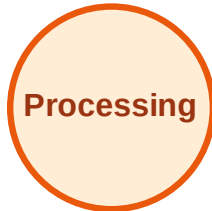
Visited: A, B, D, E, F, G



BFS Traversal

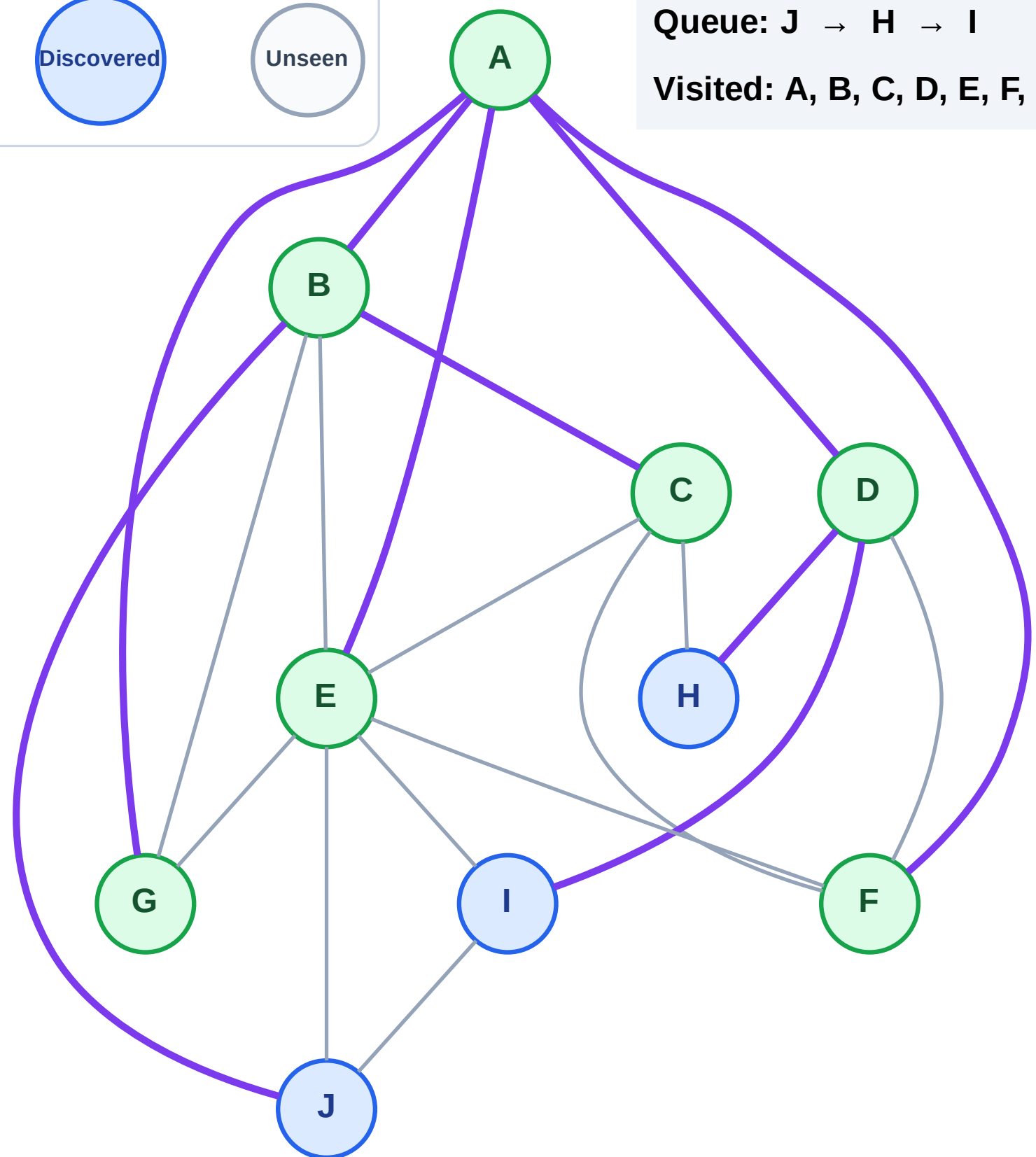
Step 15: No new neighbors from C

Legend



Queue: J → H → I

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node J

Legend

Complete

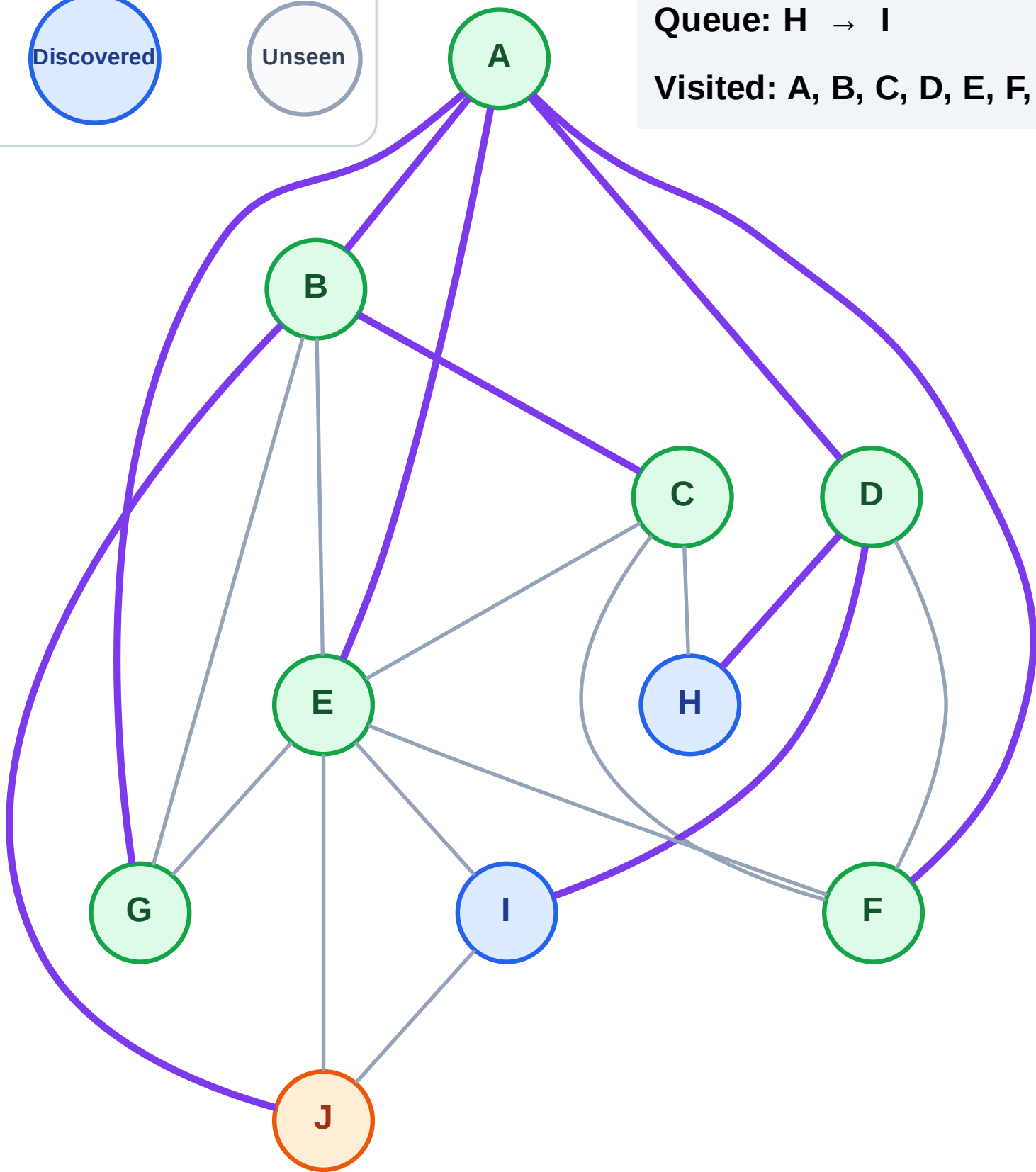
Processing

Discovered

Unseen

Queue: H → I

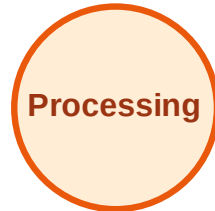
Visited: A, B, C, D, E, F, G



BFS Traversal

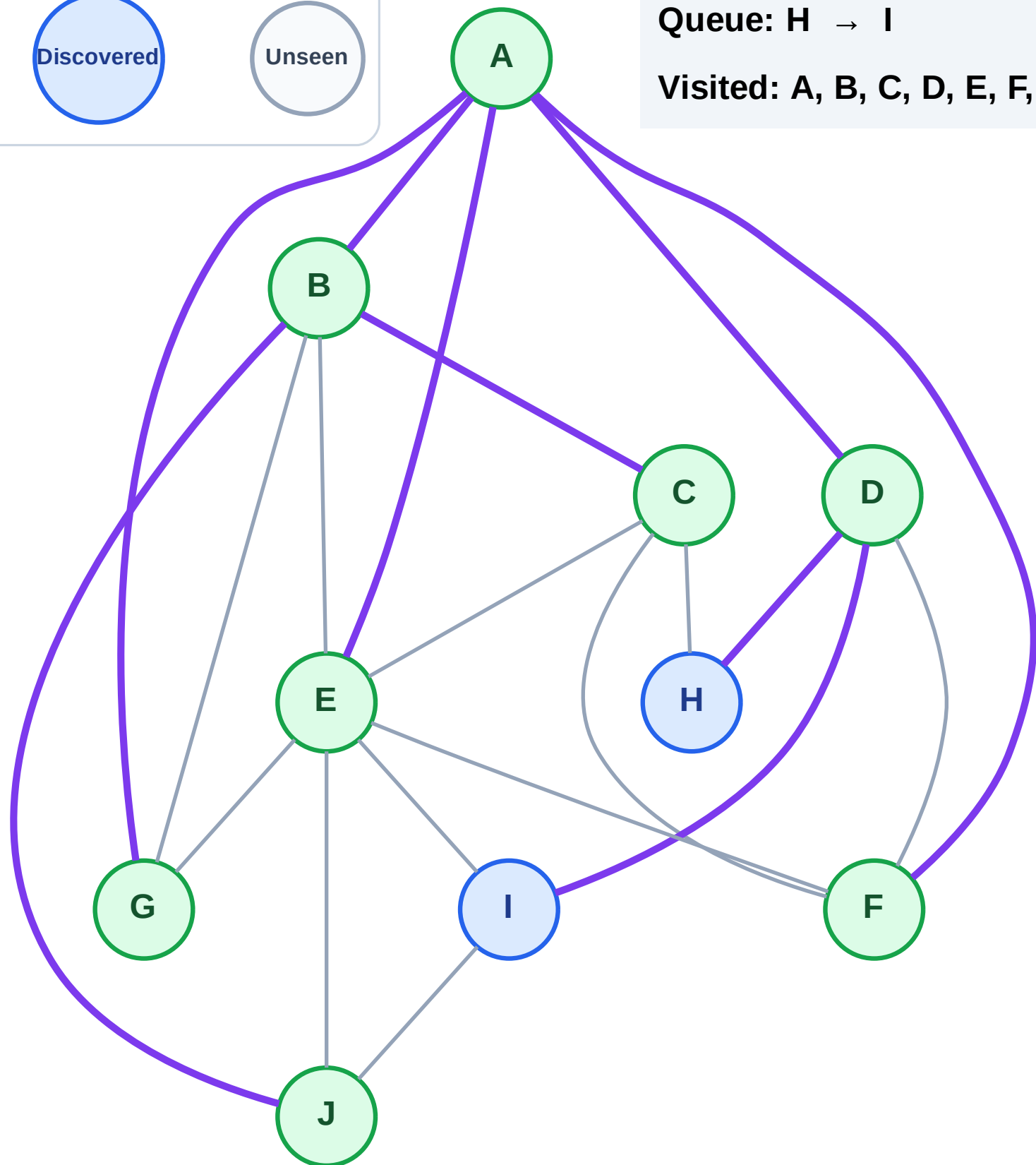
Step 17: No new neighbors from J

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

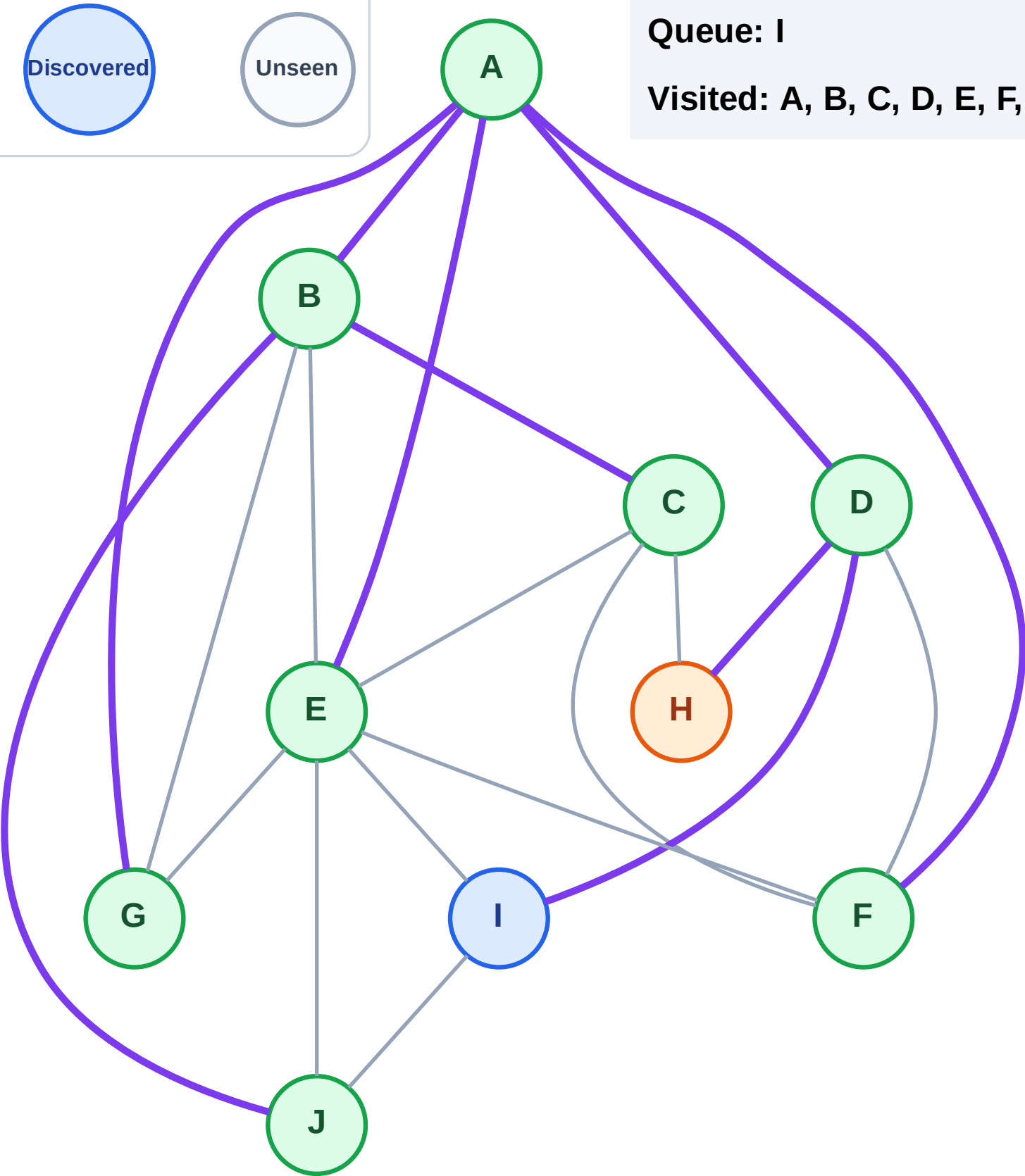
Processing

Discovered

Unseen

Queue: I

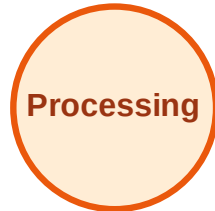
Visited: A, B, C, D, E, F, G, J



BFS Traversal

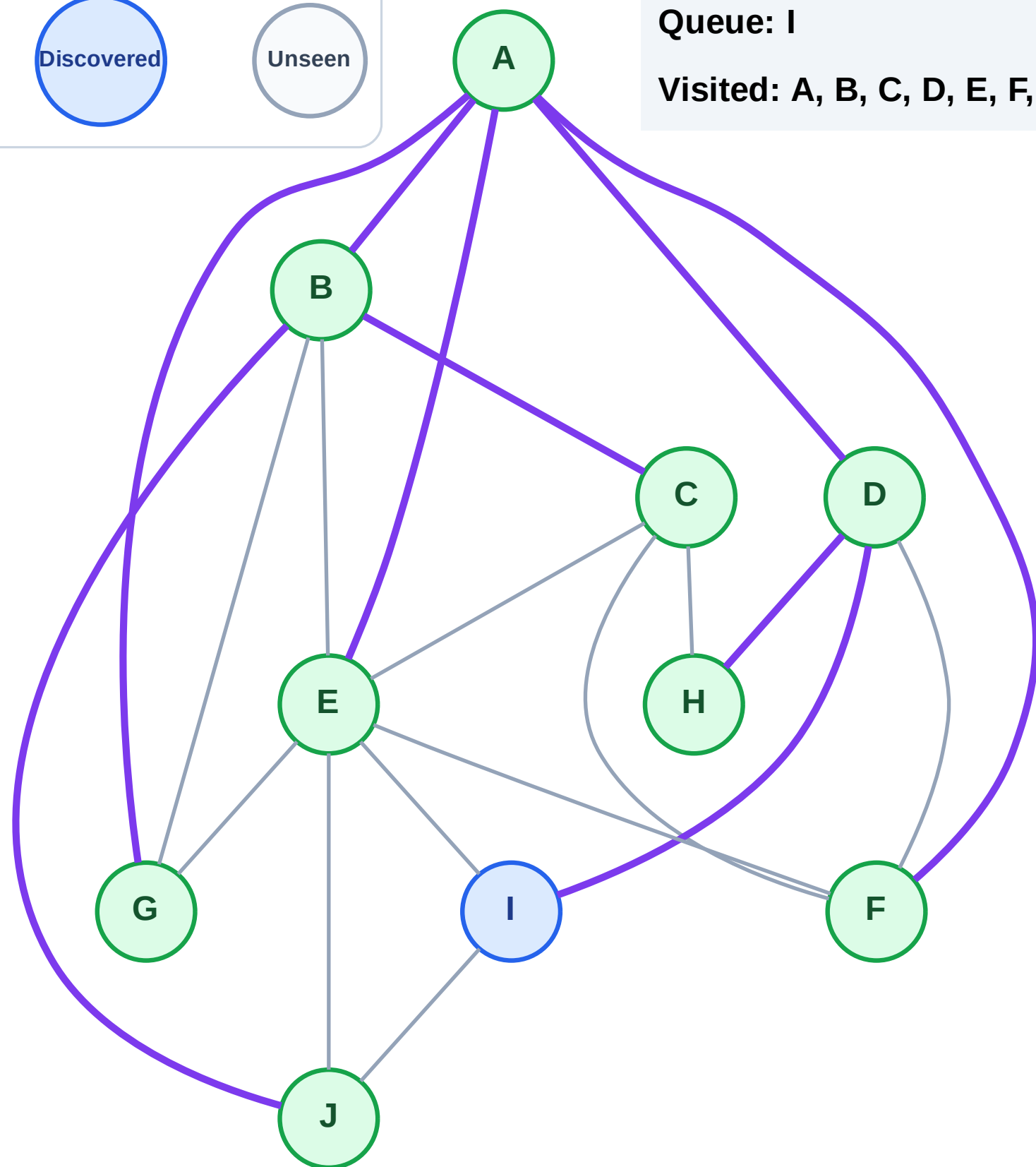
Step 19: No new neighbors from H

Legend



Queue: I

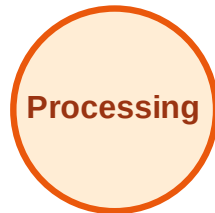
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

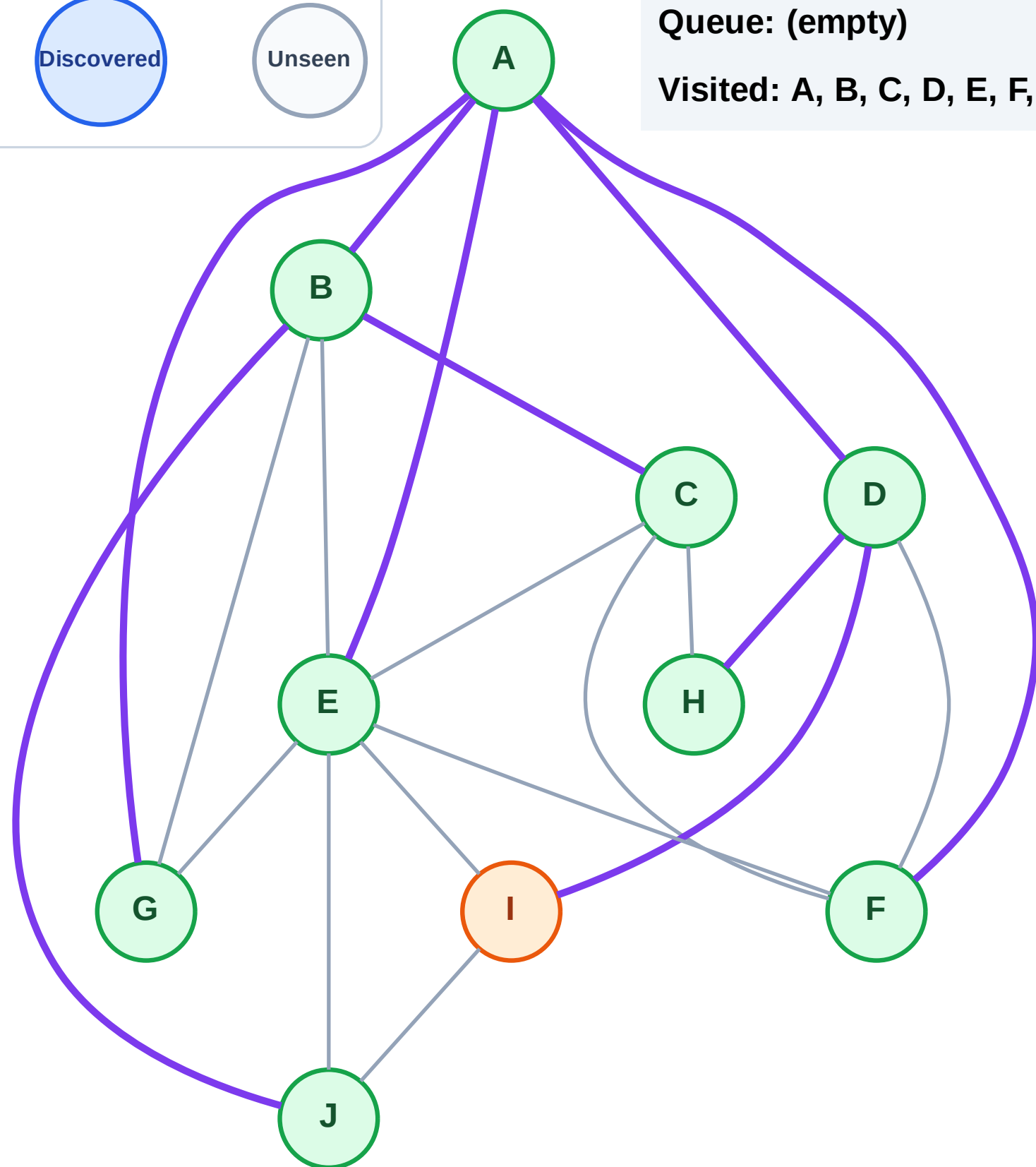
Step 20: Process node I

Legend



Queue: (empty)

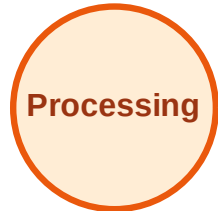
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

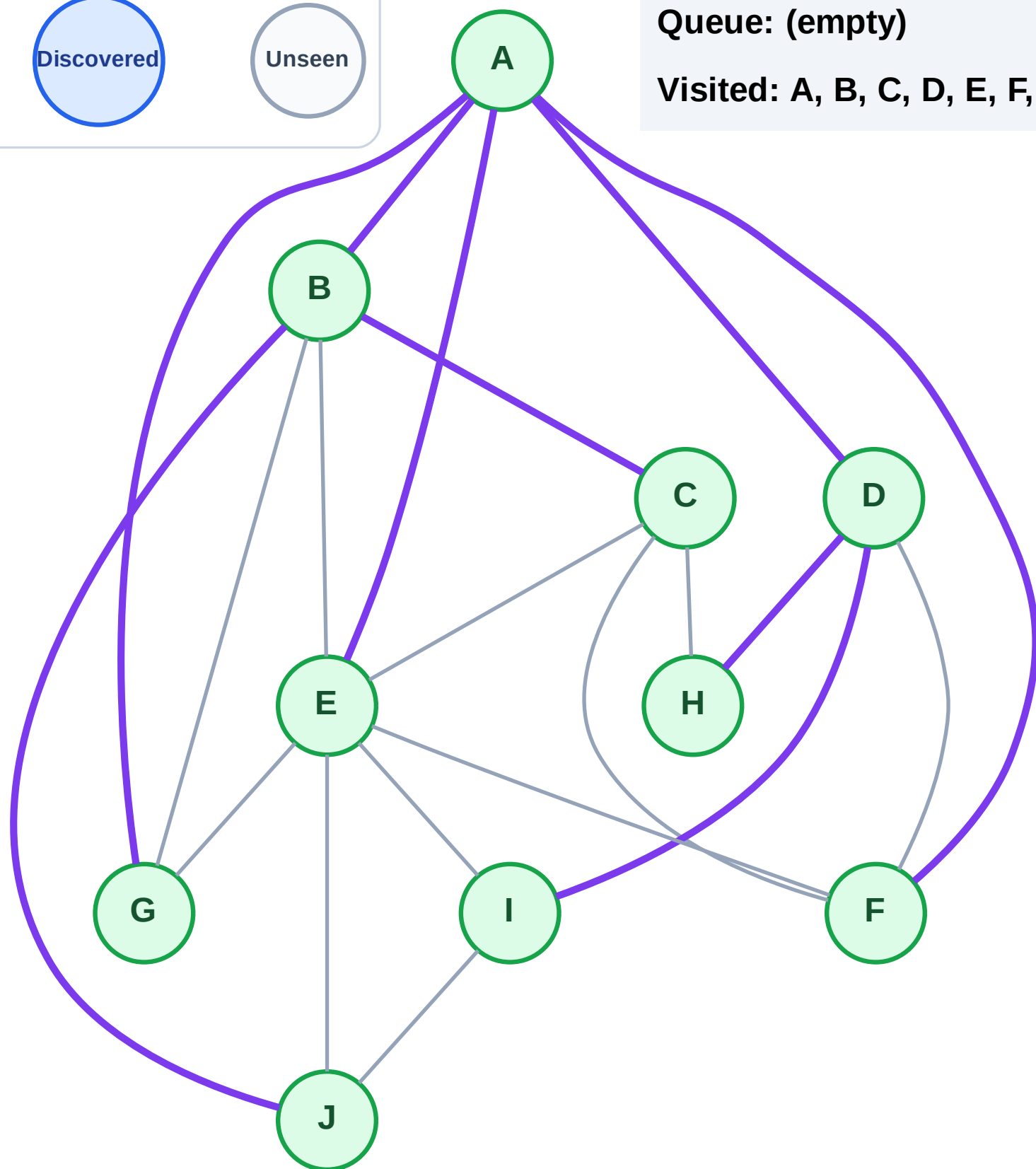
Step 21: No new neighbors from I

Legend



Queue: (empty)

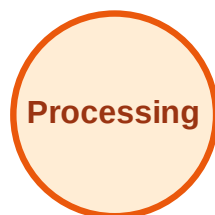
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

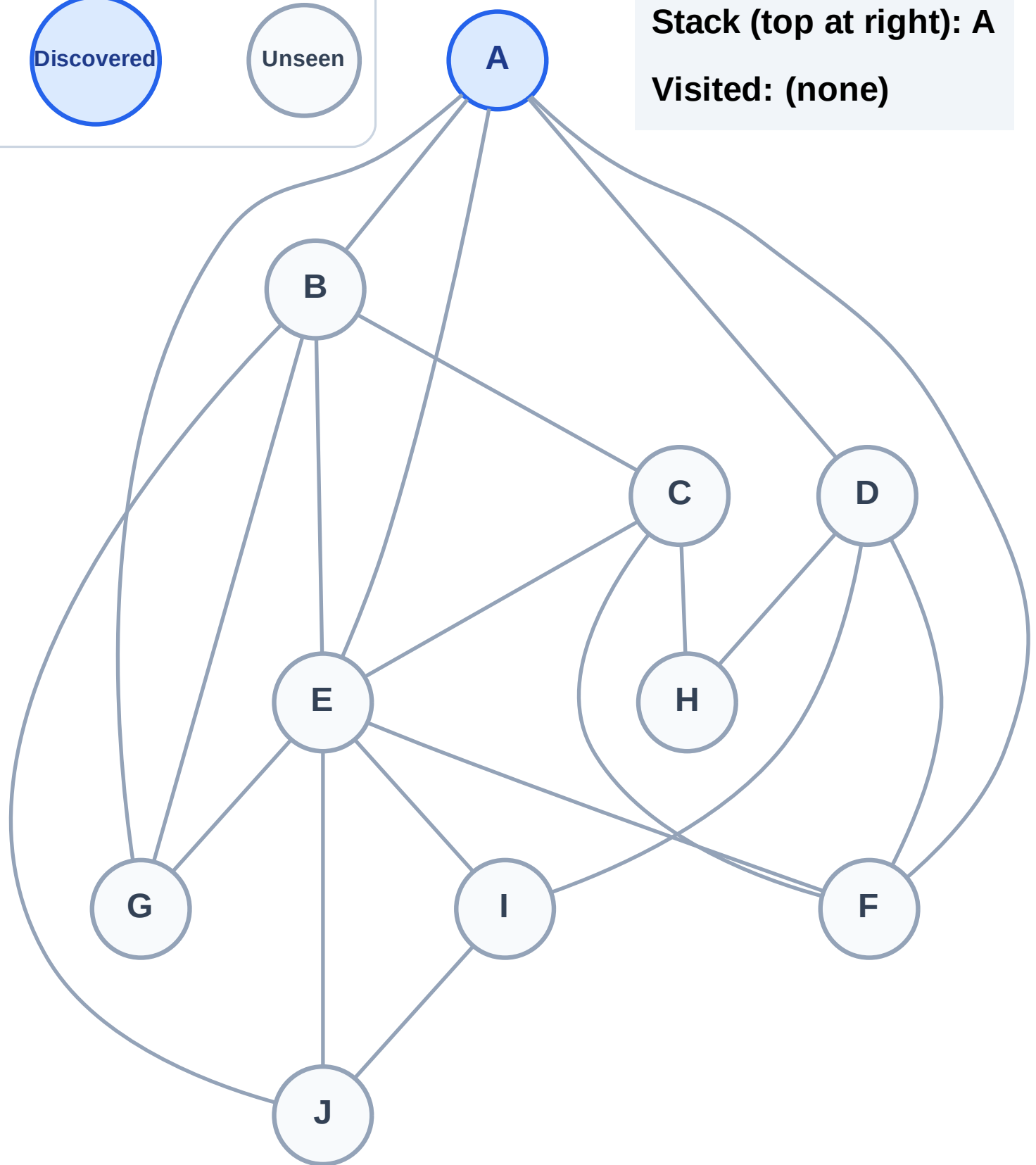
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

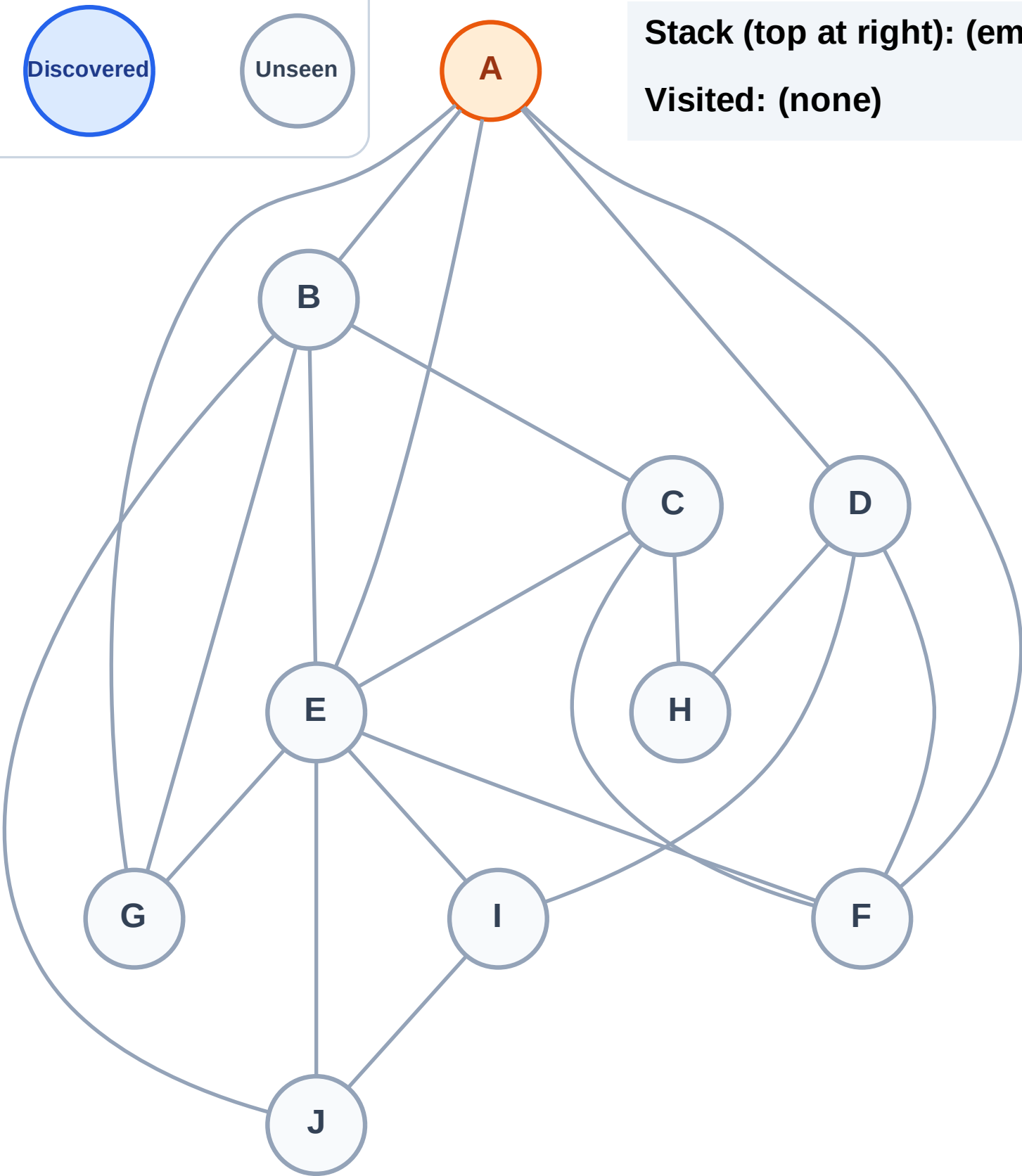
Complete

Processing

Discovered

Unseen

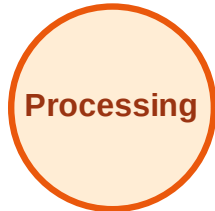
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

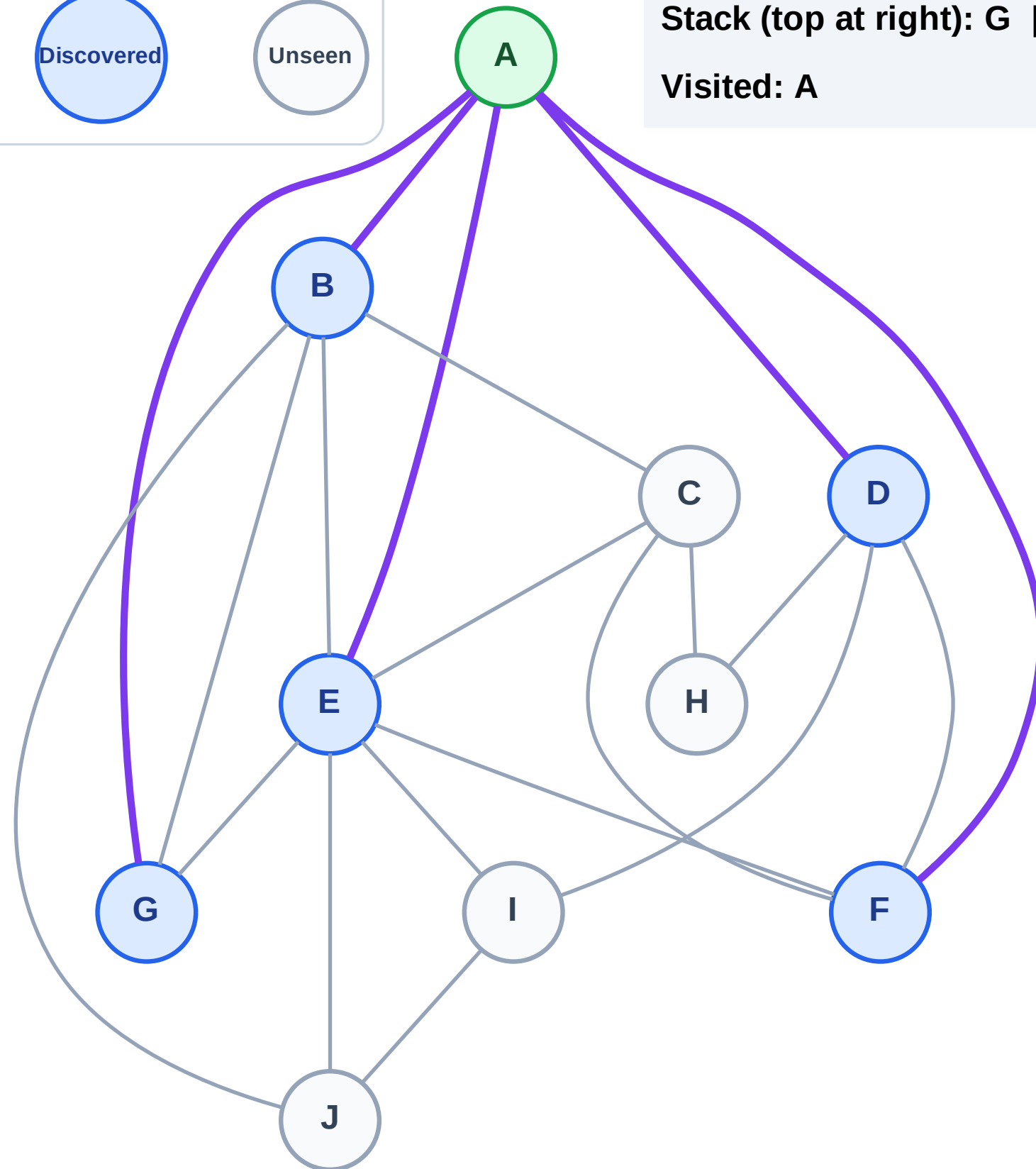
Step 3: Discover G, F, E, D, B from A

Legend



Stack (top at right): G | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

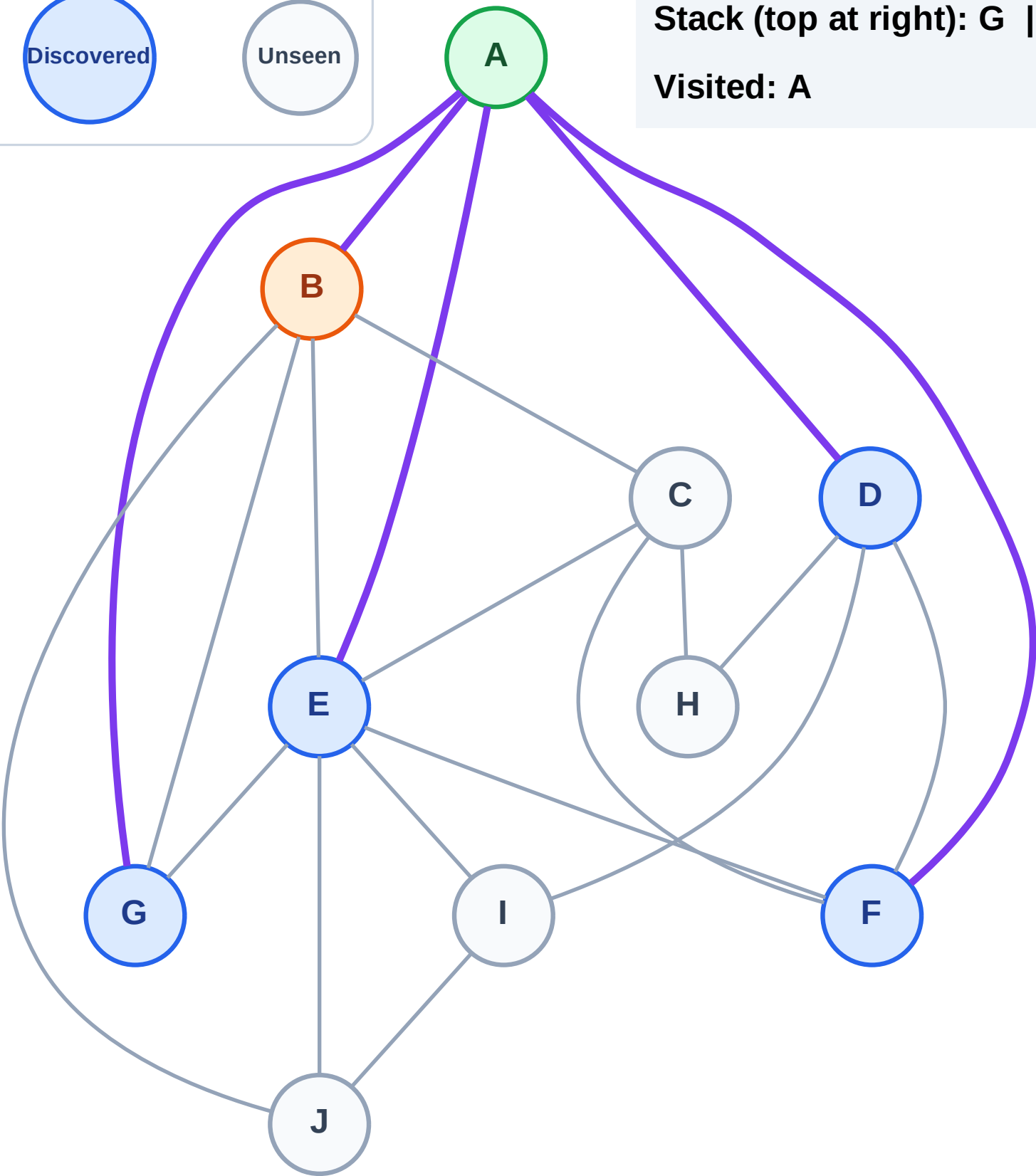
Complete

Processing

Discovered

Unseen

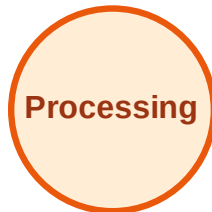
Stack (top at right): G | F | E | D
Visited: A



DFS Traversal

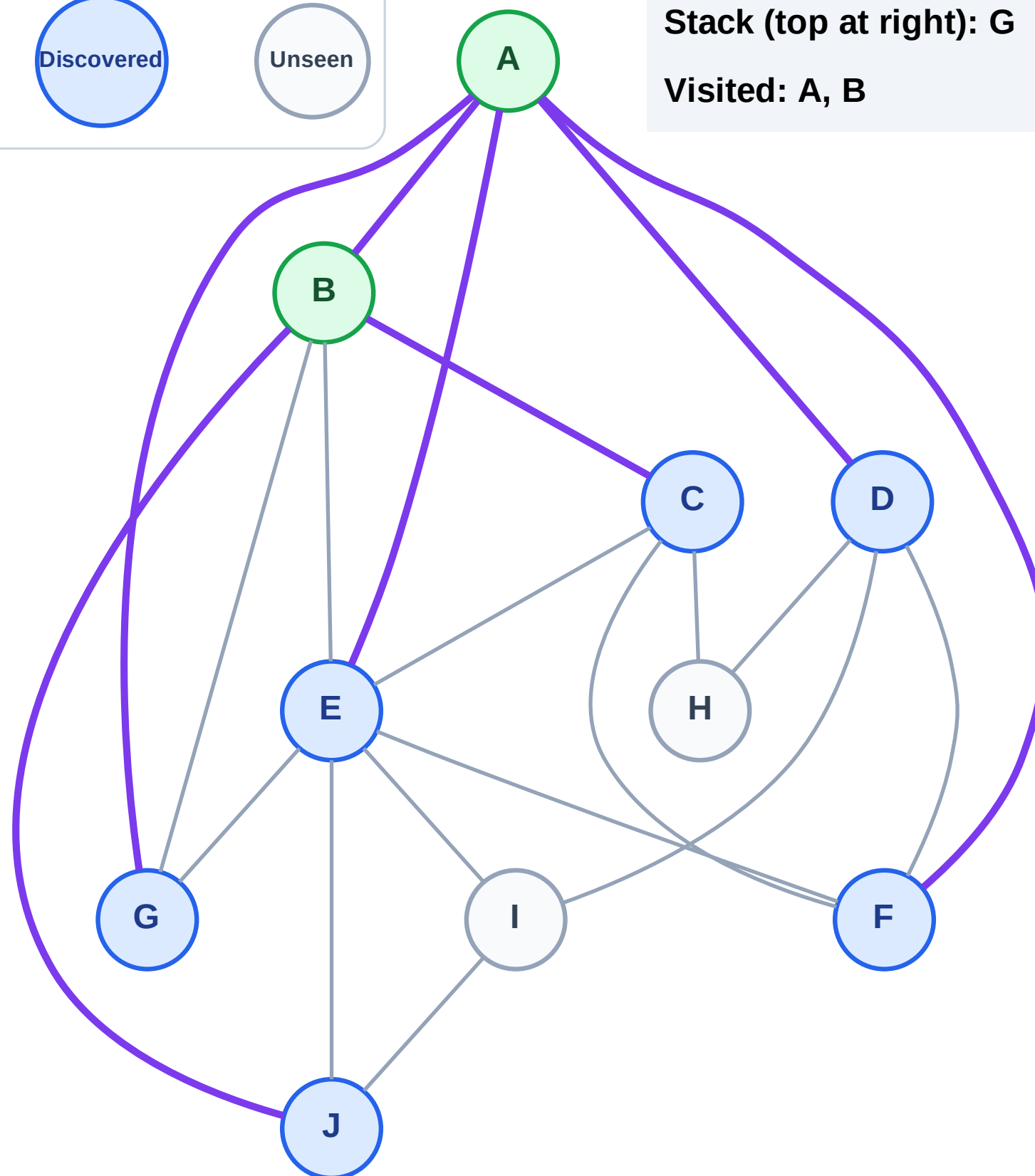
Step 5: Discover J, C from B

Legend



Stack (top at right): G | F | E | D | J | C

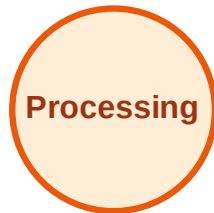
Visited: A, B



DFS Traversal

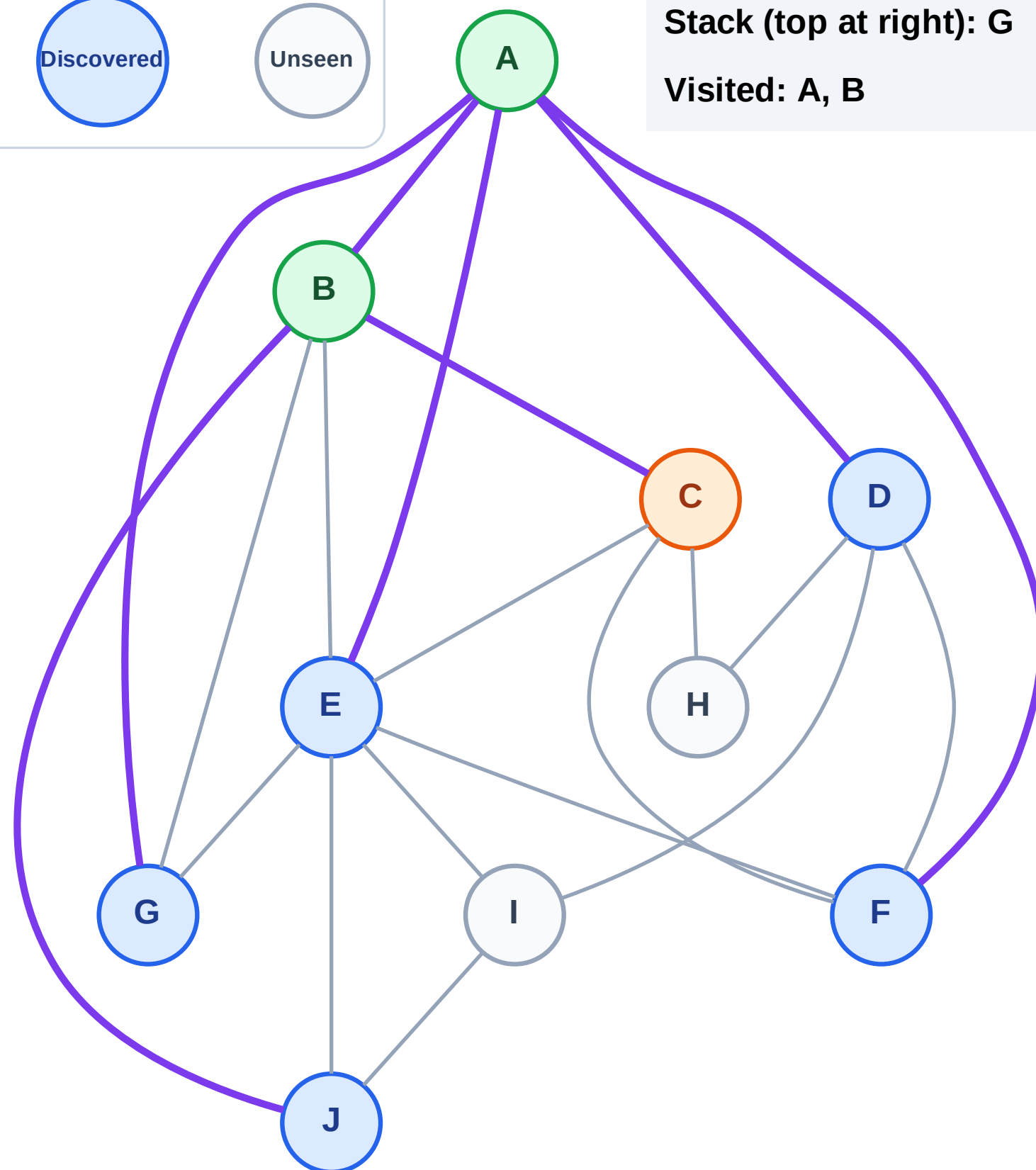
Step 6: Process node C

Legend



Stack (top at right): G | F | E | D | J

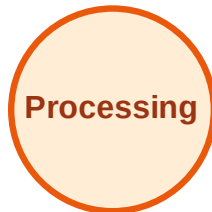
Visited: A, B



DFS Traversal

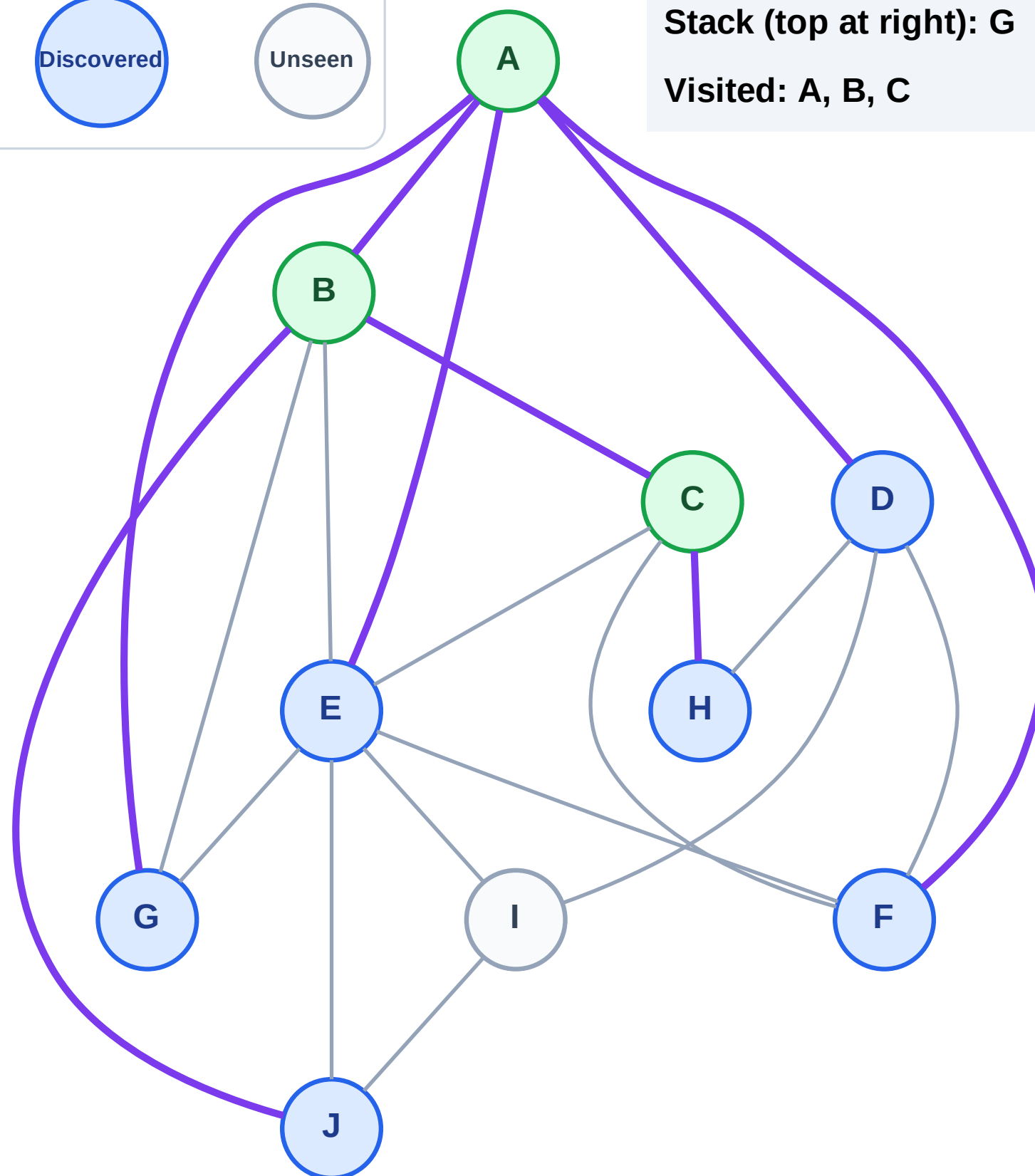
Step 7: Discover H from C

Legend



Stack (top at right): G | F | E | D | J | H

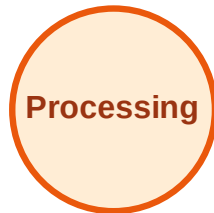
Visited: A, B, C



DFS Traversal

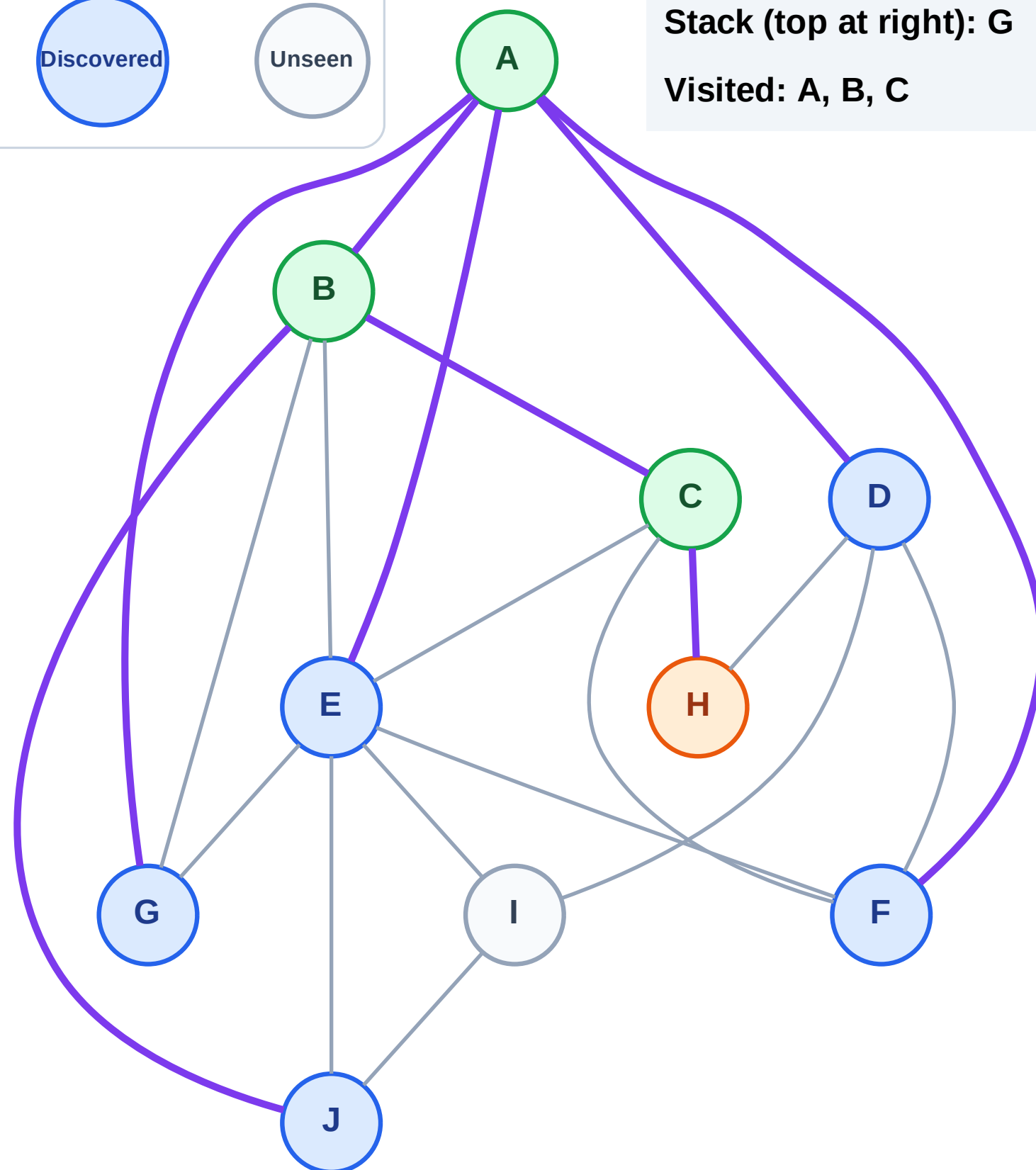
Step 8: Process node H

Legend



Stack (top at right): G | F | E | D | J

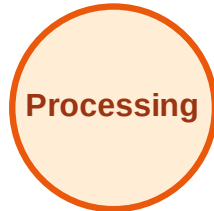
Visited: A, B, C



DFS Traversal

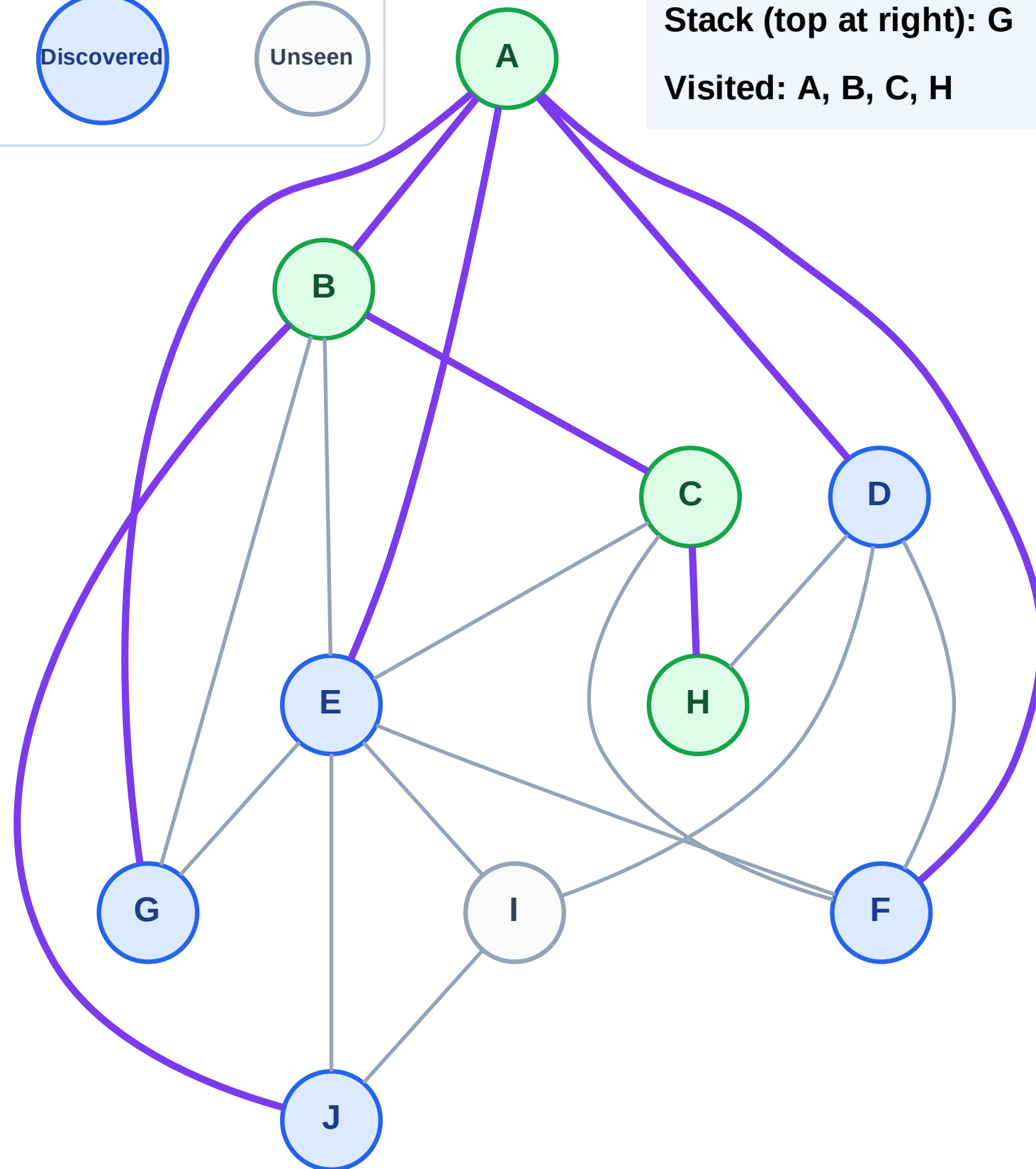
Step 9: No new neighbors from H

Legend



Stack (top at right): G | F | E | D | J

Visited: A, B, C, H



DFS Traversal

Step 10: Process node J

Legend

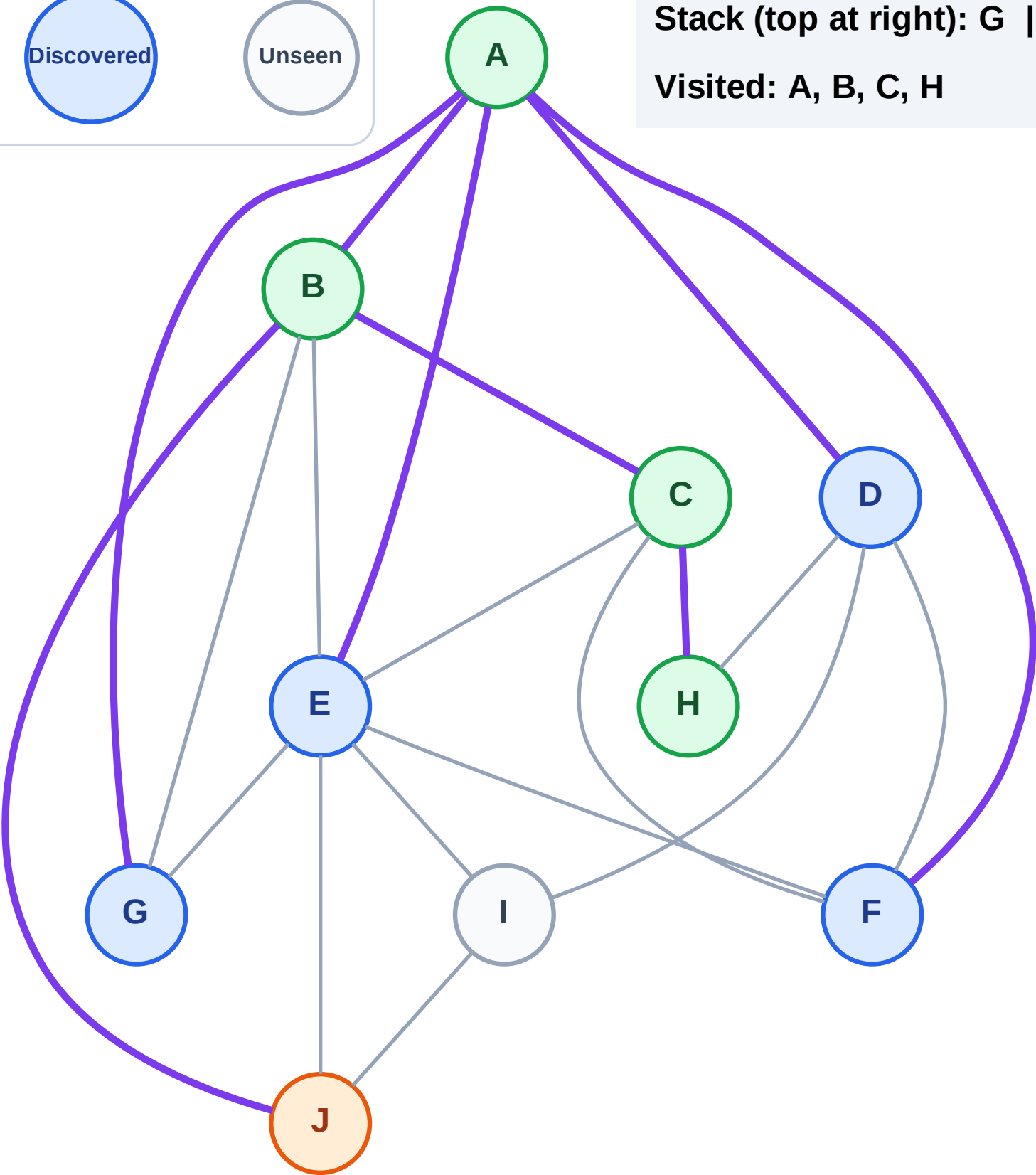
Complete

Processing

Discovered

Unseen

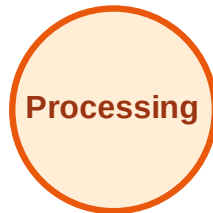
Stack (top at right): G | F | E | D
Visited: A, B, C, H



DFS Traversal

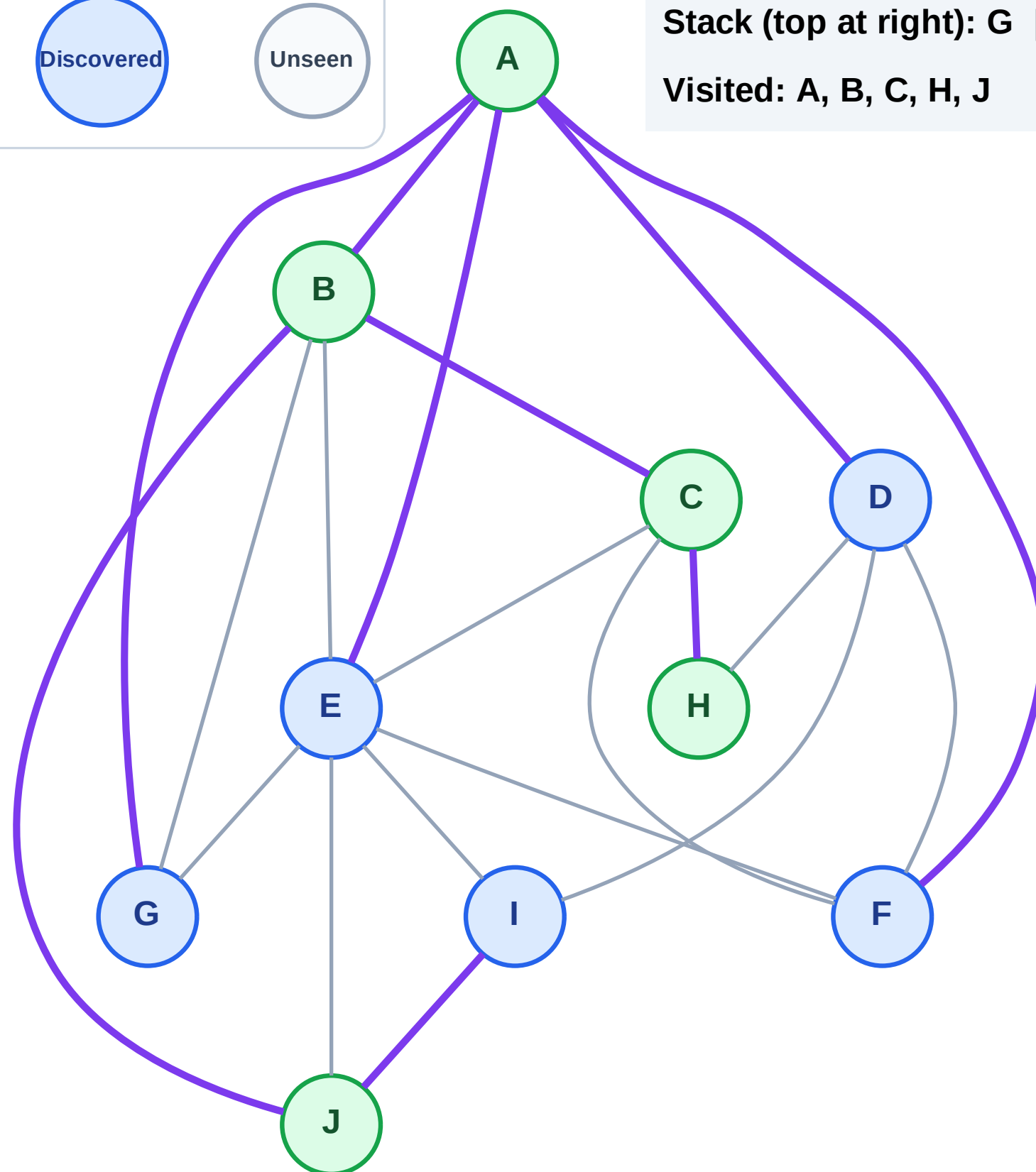
Step 11: Discover I from J

Legend



Stack (top at right): G | F | E | D | I

Visited: A, B, C, H, J



DFS Traversal

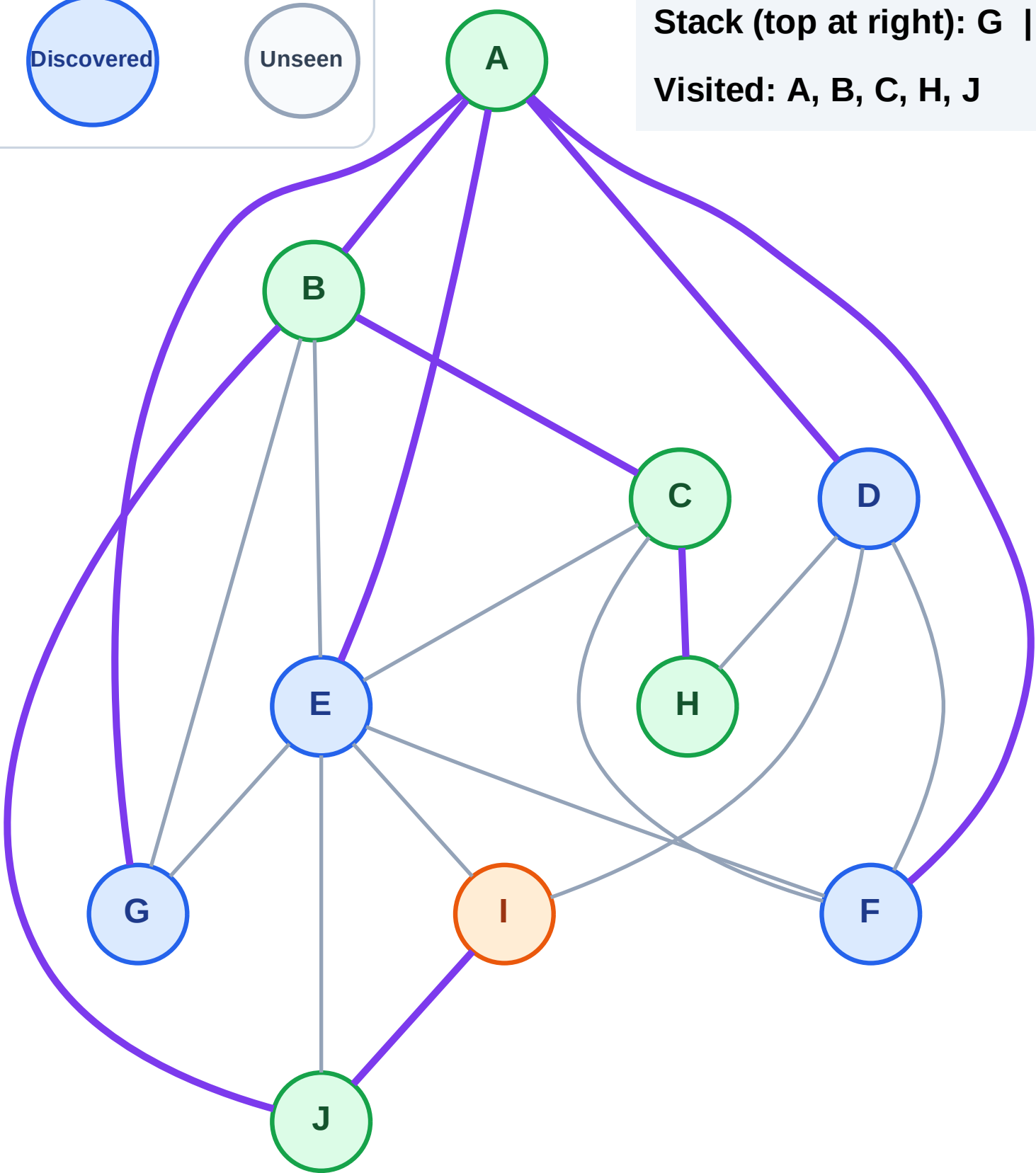
Step 12: Process node I

Legend



Stack (top at right): G | F | E | D

Visited: A, B, C, H, J



DFS Traversal

Step 13: No new neighbors from I

Legend

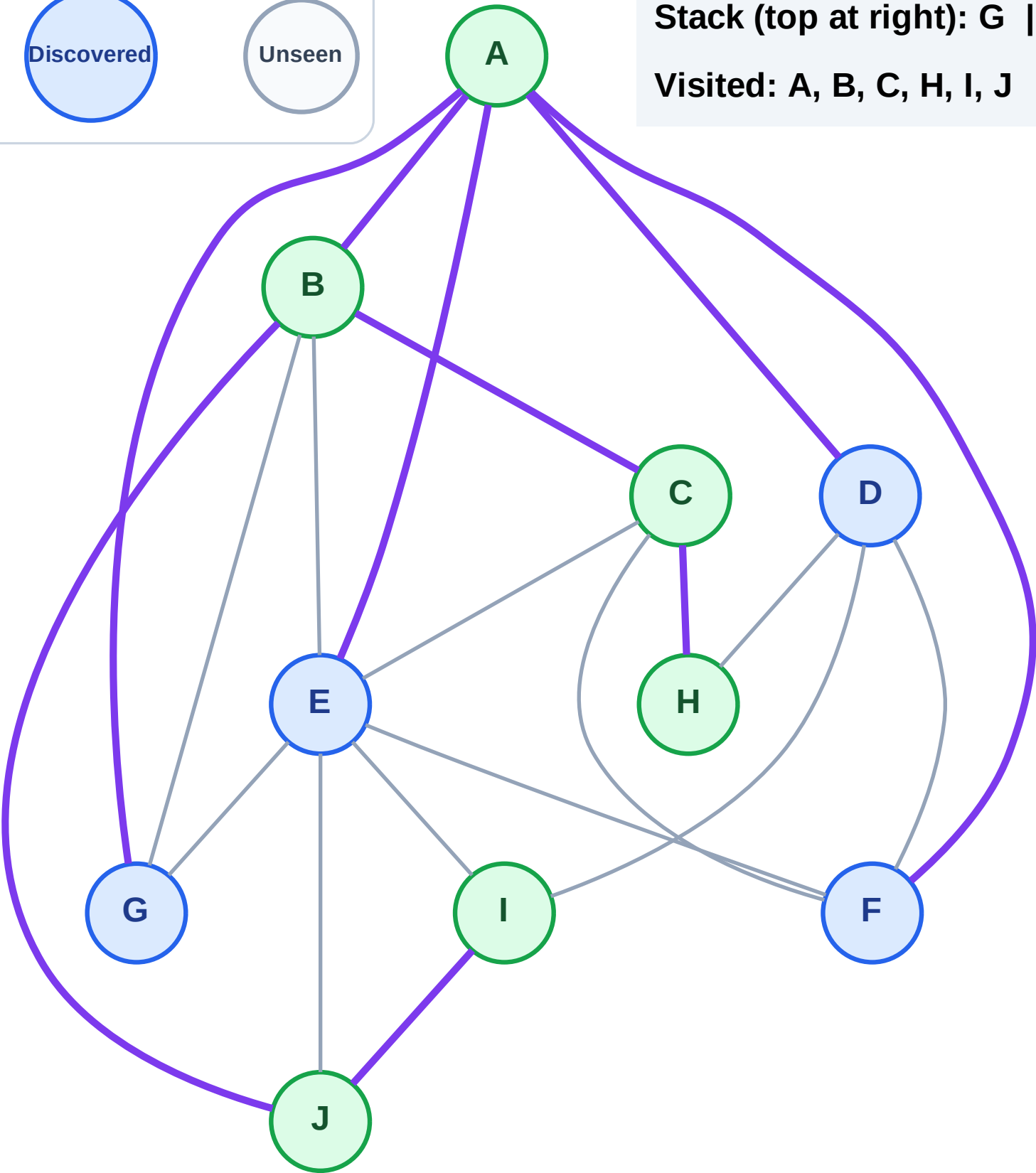
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | E | D
Visited: A, B, C, H, I, J

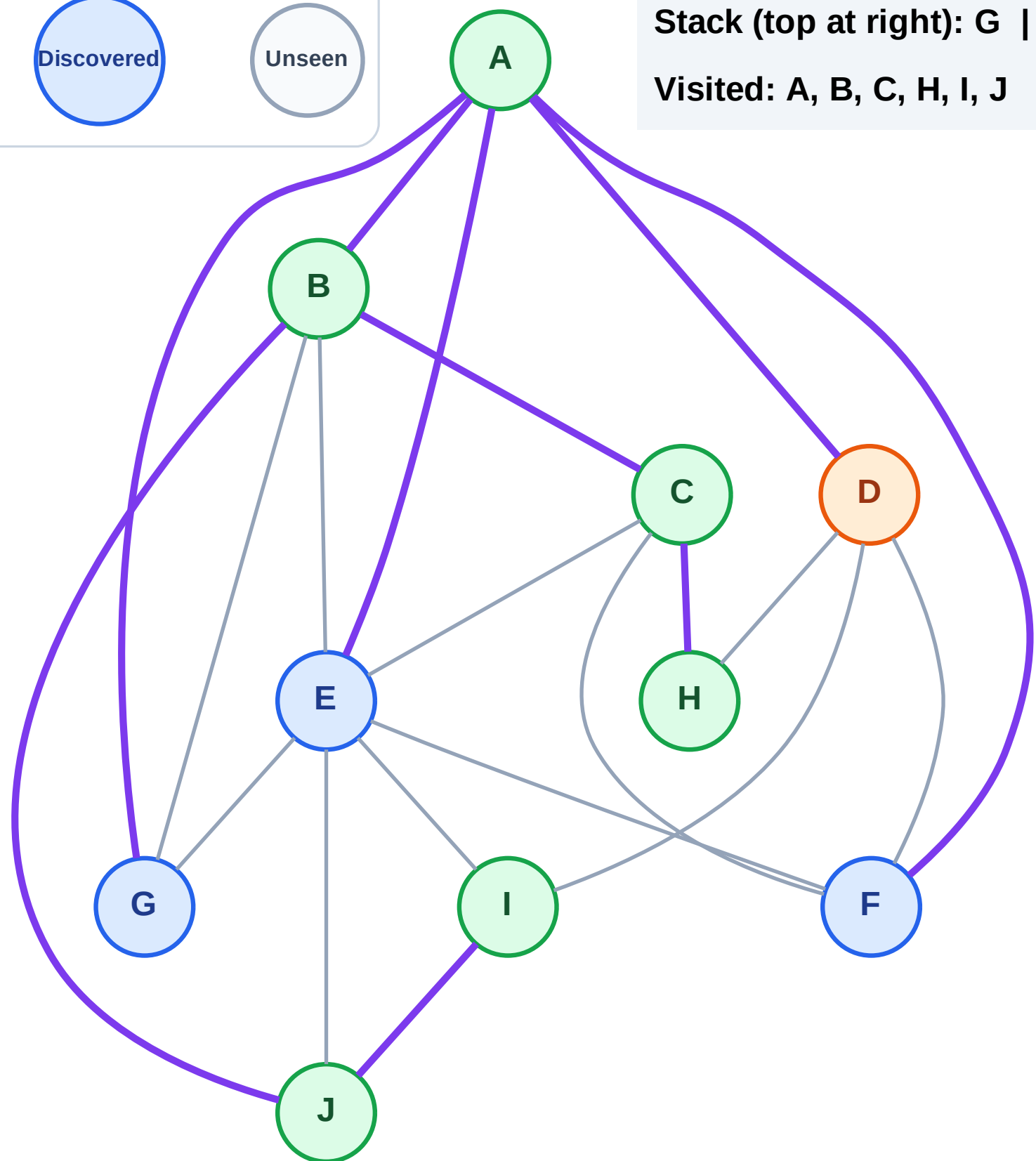


Step 14: Process node D

Legend



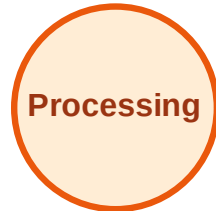
Visited: A, B, C, H, I, J



DFS Traversal

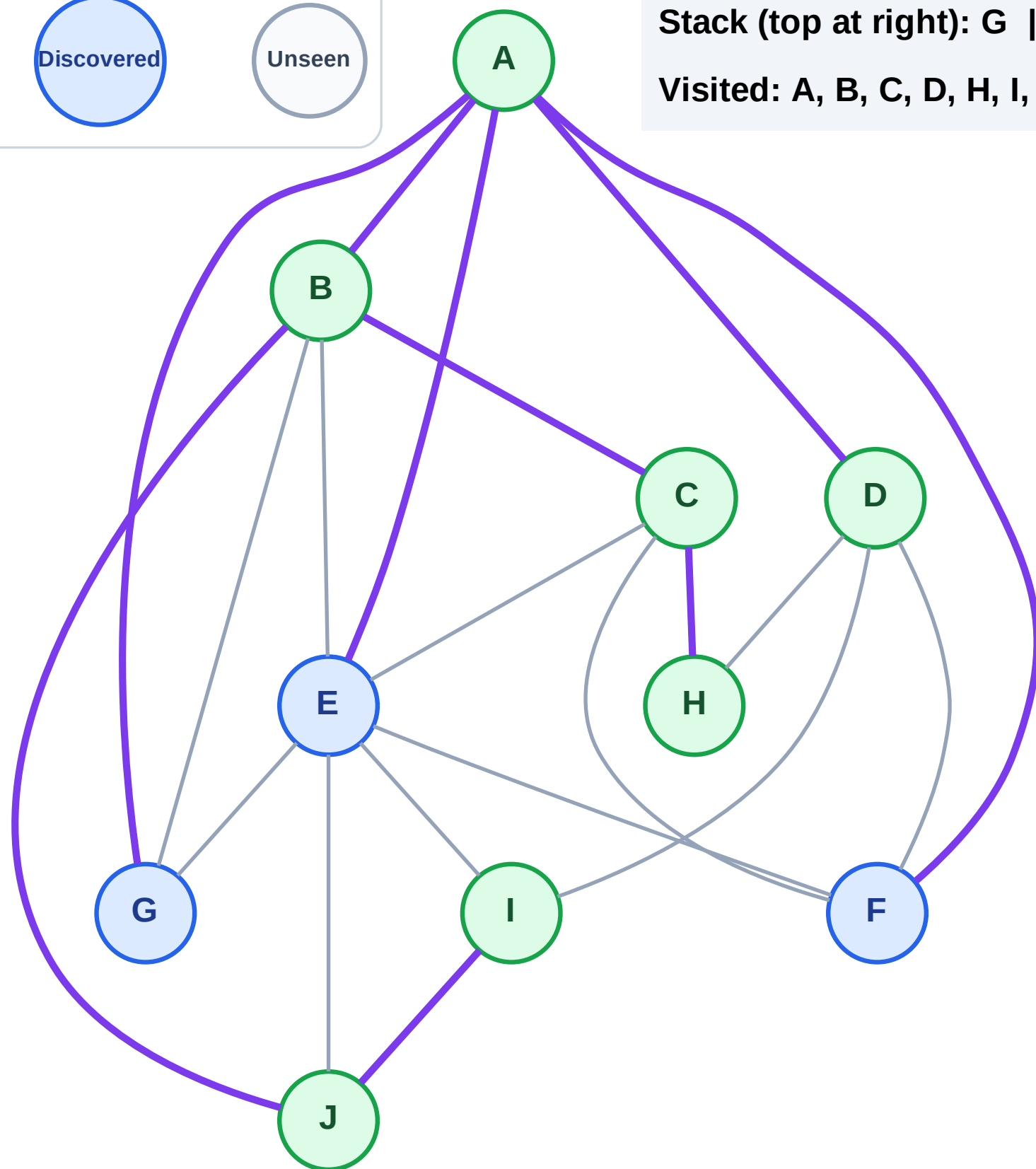
Step 15: No new neighbors from D

Legend



Stack (top at right): G | F | E

Visited: A, B, C, D, H, I, J



DFS Traversal

Step 16: Process node E

Legend

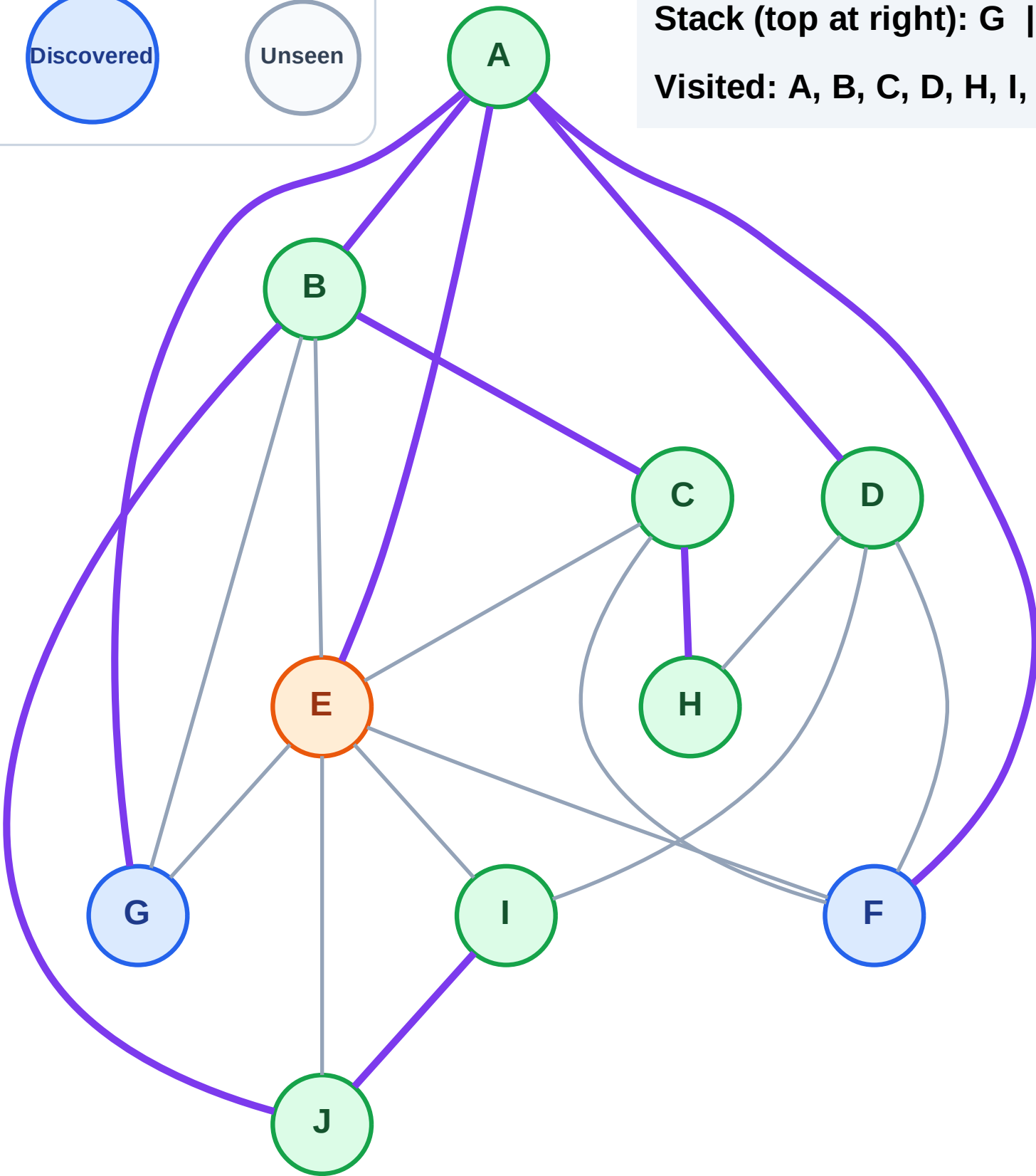
Complete

Processing

Discovered

Unseen

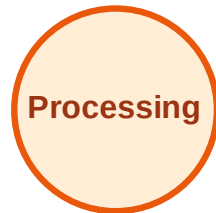
Stack (top at right): G | F
Visited: A, B, C, D, H, I, J



DFS Traversal

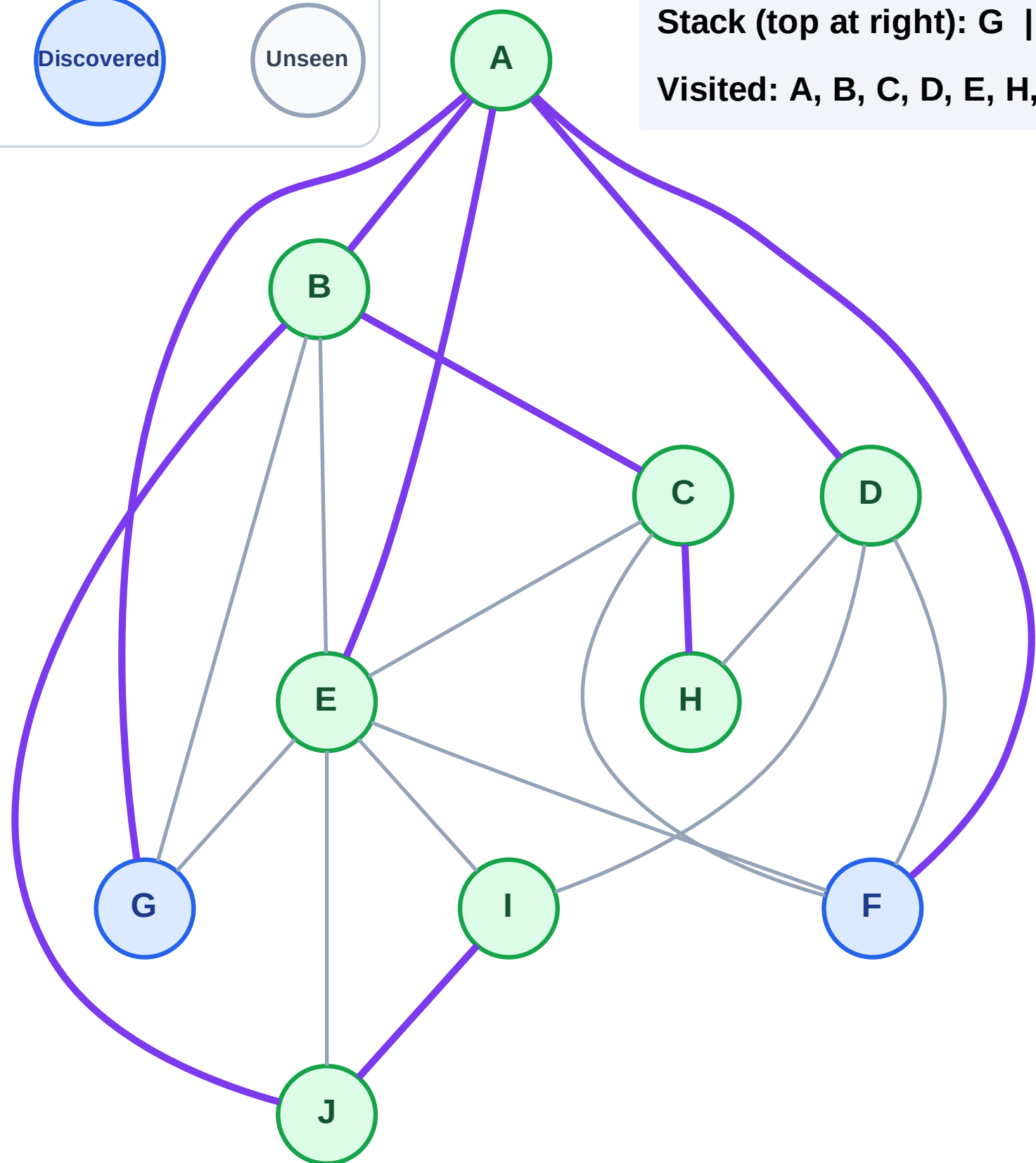
Step 17: No new neighbors from E

Legend



Stack (top at right): G | F

Visited: A, B, C, D, E, H, I, J



DFS Traversal

Step 18: Process node F

Legend

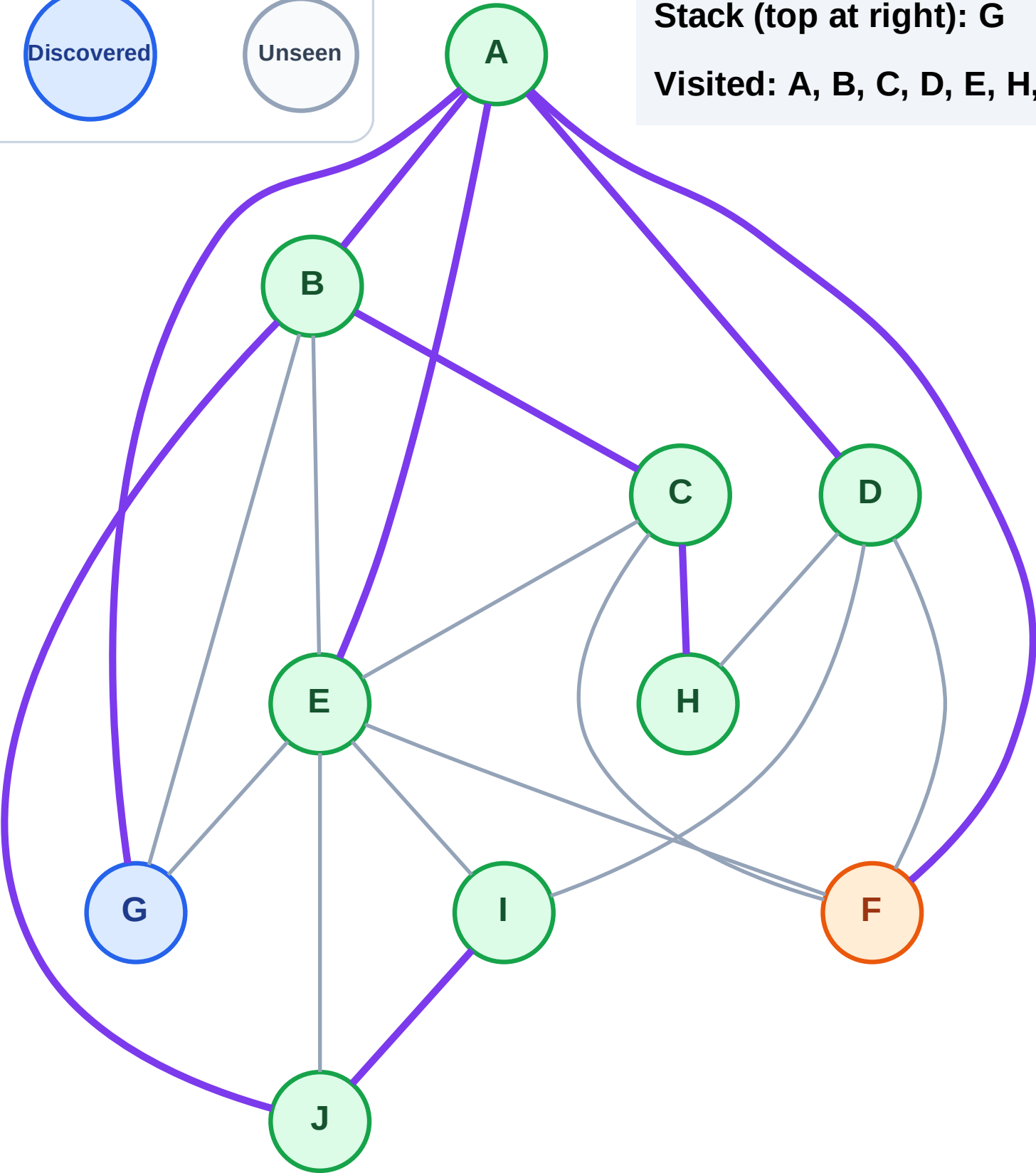
Complete

Processing

Discovered

Unseen

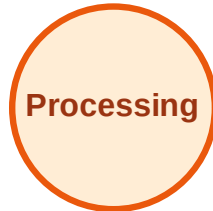
Stack (top at right): G
Visited: A, B, C, D, E, H, I, J



DFS Traversal

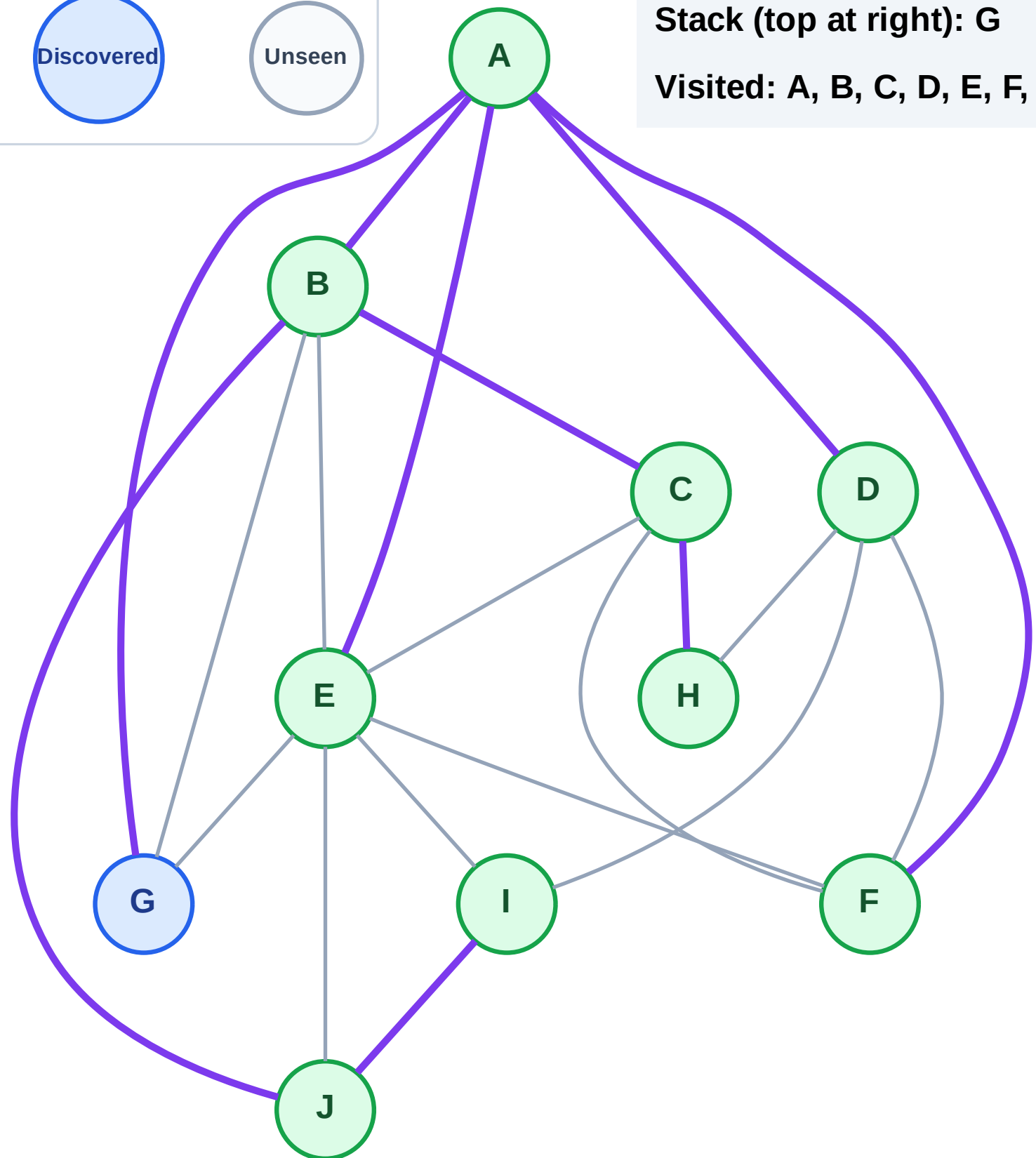
Step 19: No new neighbors from F

Legend



Stack (top at right): G

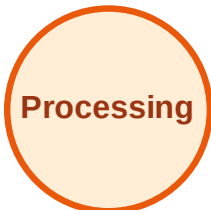
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

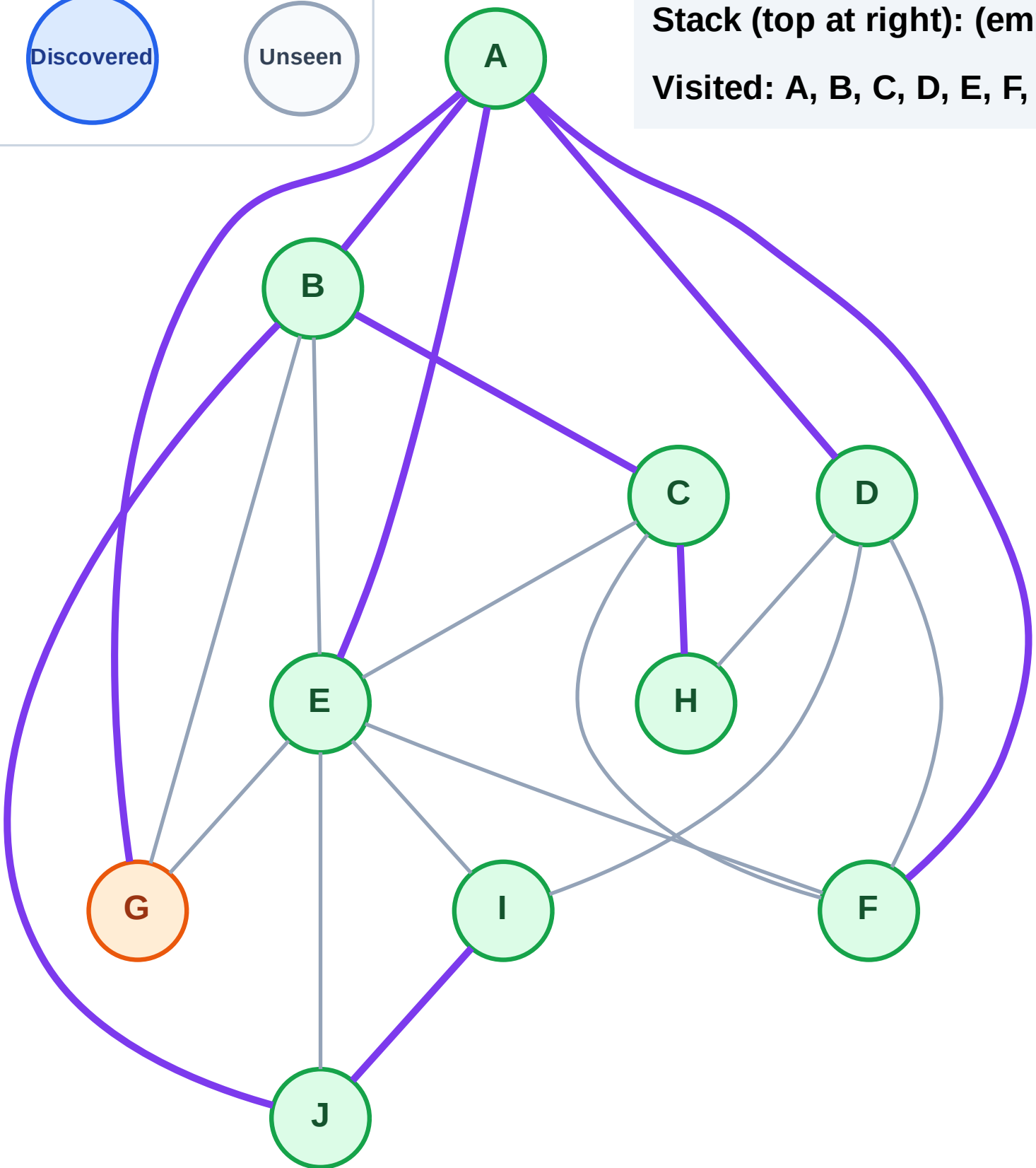
Step 20: Process node G

Legend



Stack (top at right): (empty)

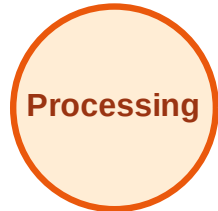
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

